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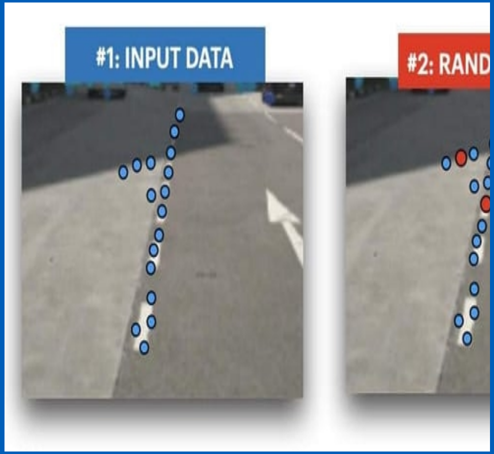
Model Fitting



Summary

- What is Model Fitting?
- Is it difficult?
- Least Squares (LS)
  - Line fitting example
  - LS issues (outliers)
- Robust Statistics
- RANSAC

Model Fitting

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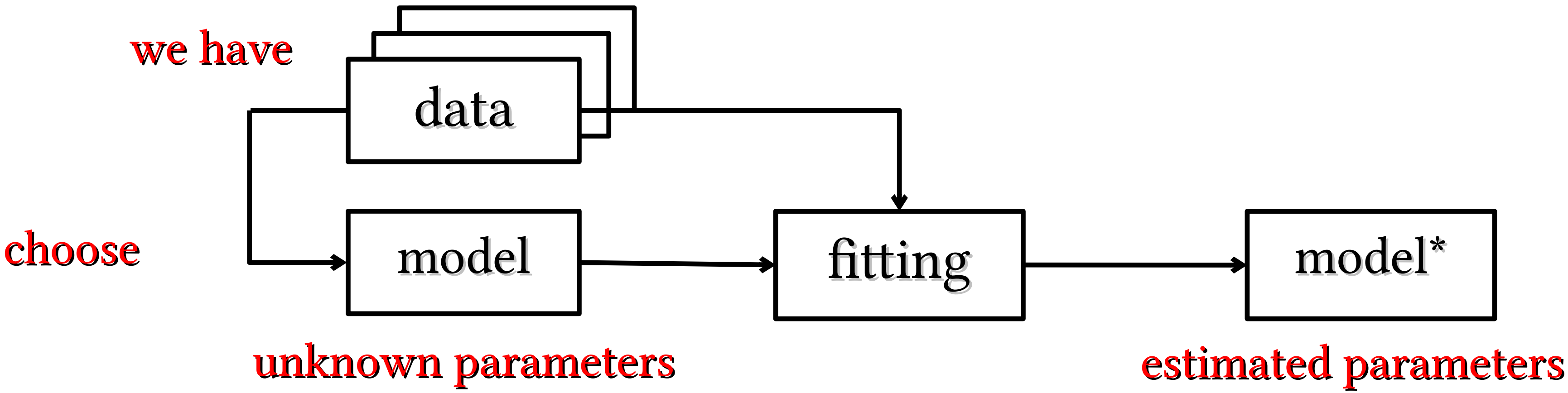
- We have data
  - A lot of them
- Is there a model that generalize data distribution?
  - Choose a model
  - Compute model parameters

Model Fitting



Models

- Lines, curves, planes…
- Homographies, Calibration Matrices…
- …

Example: lines fitting

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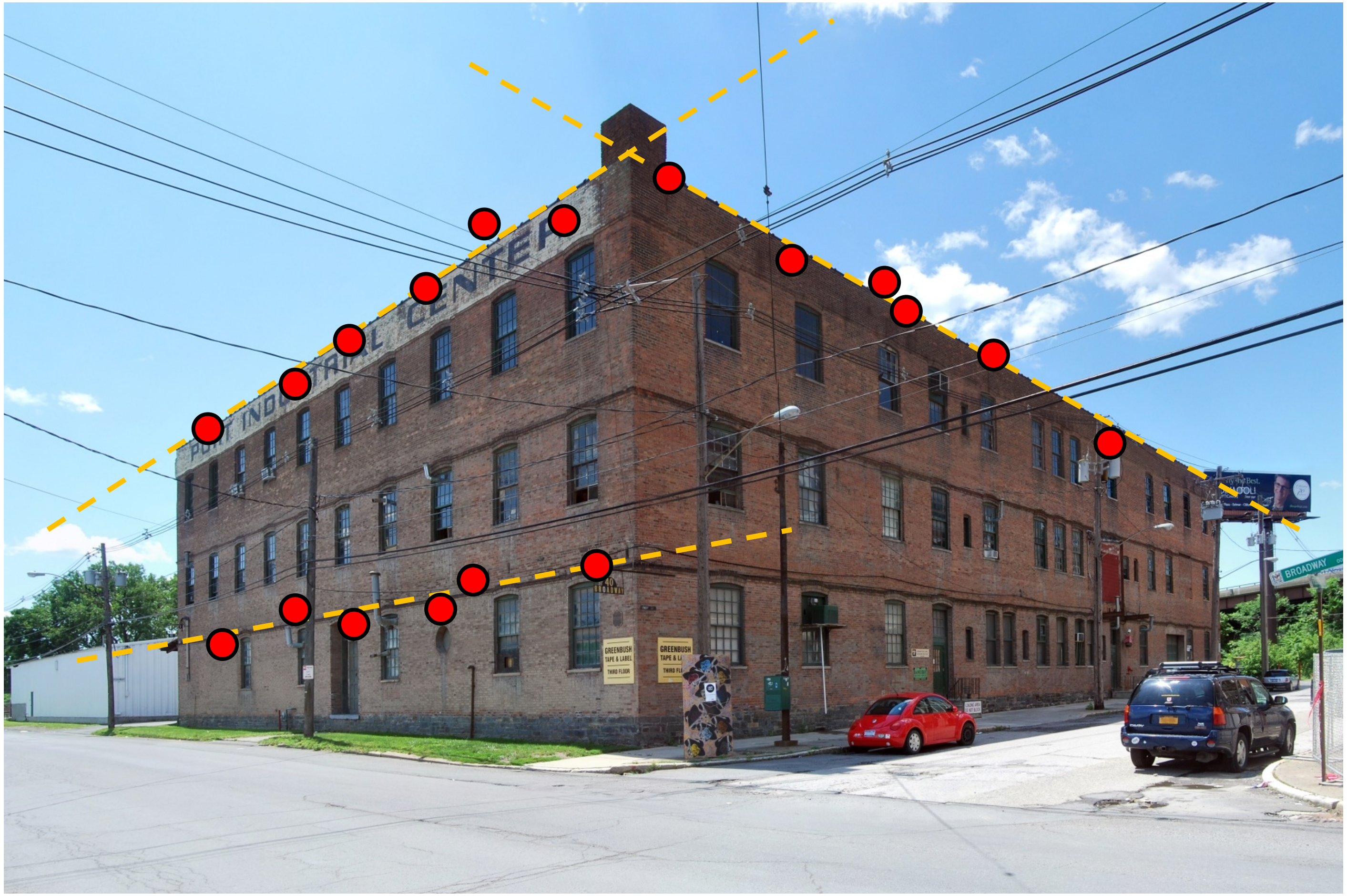
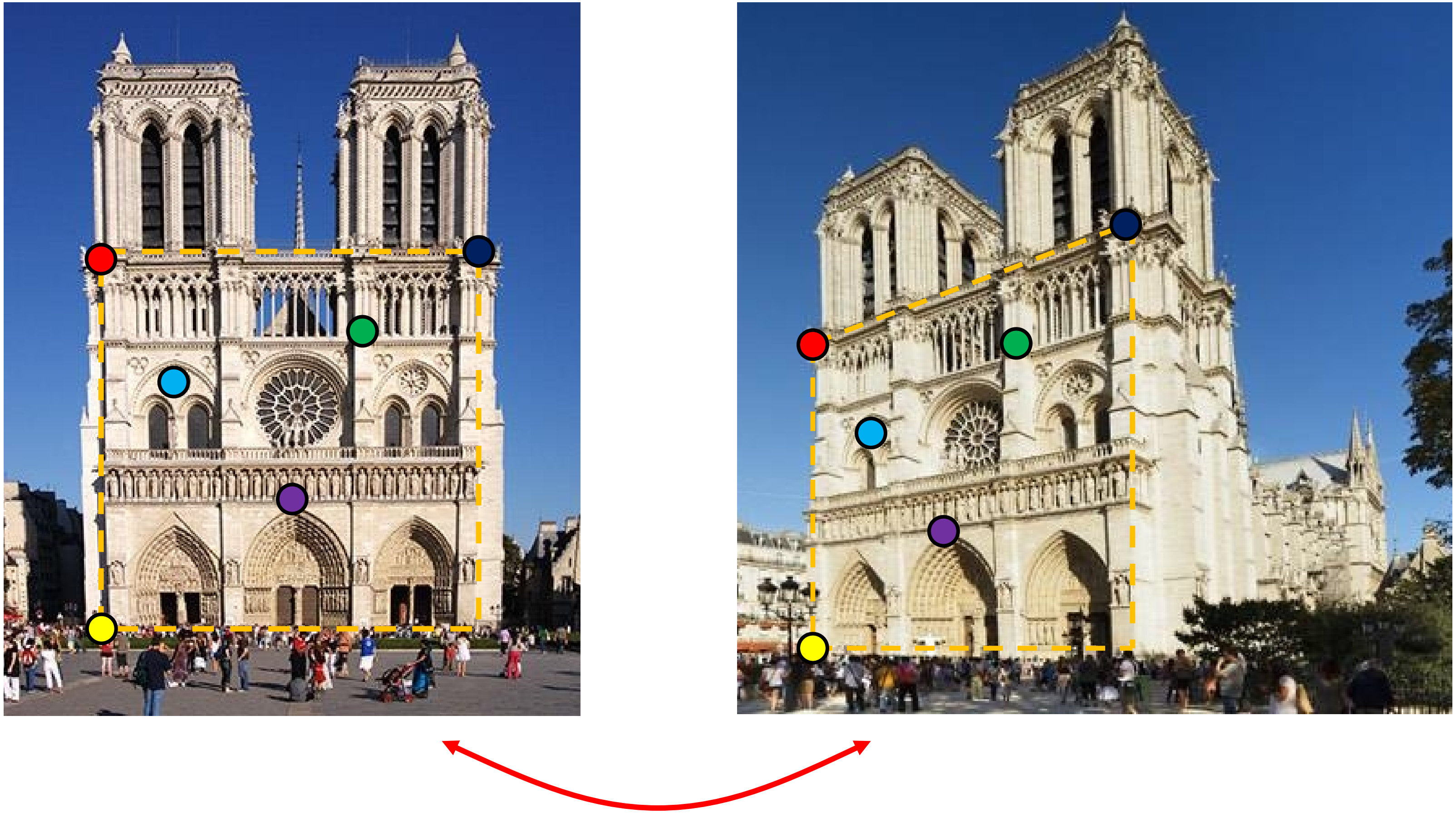


Example: homograpy



Problems

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- Noise
- Outliers
- Missing data

Outliers

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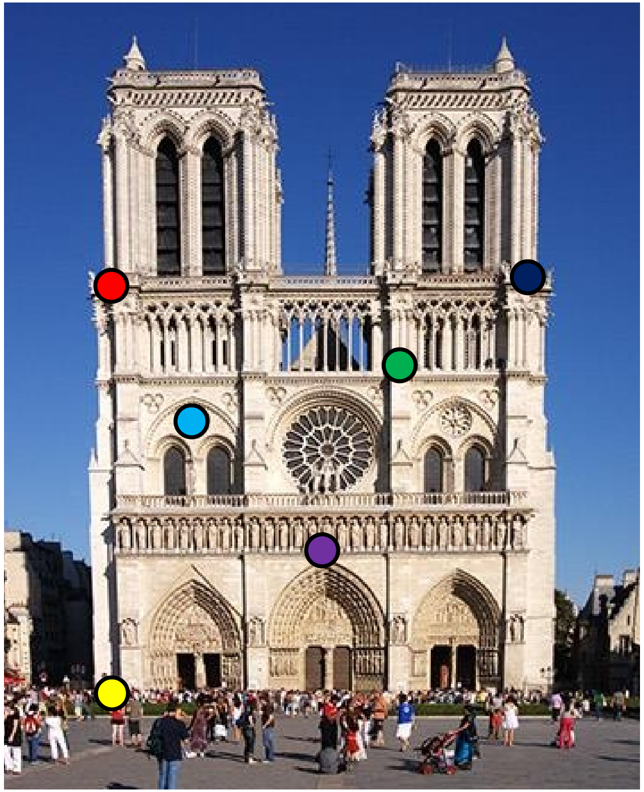
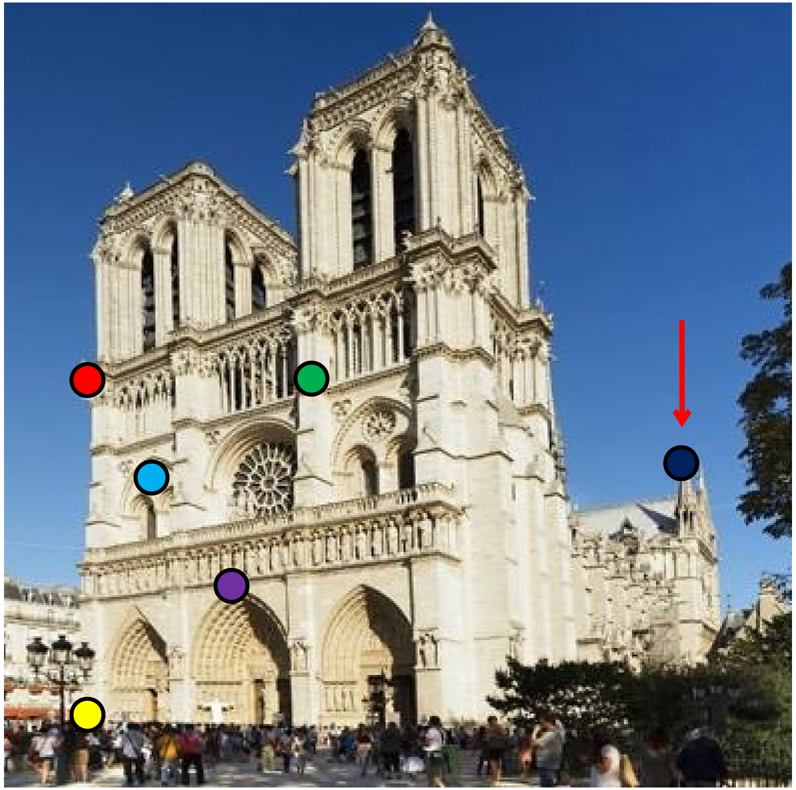
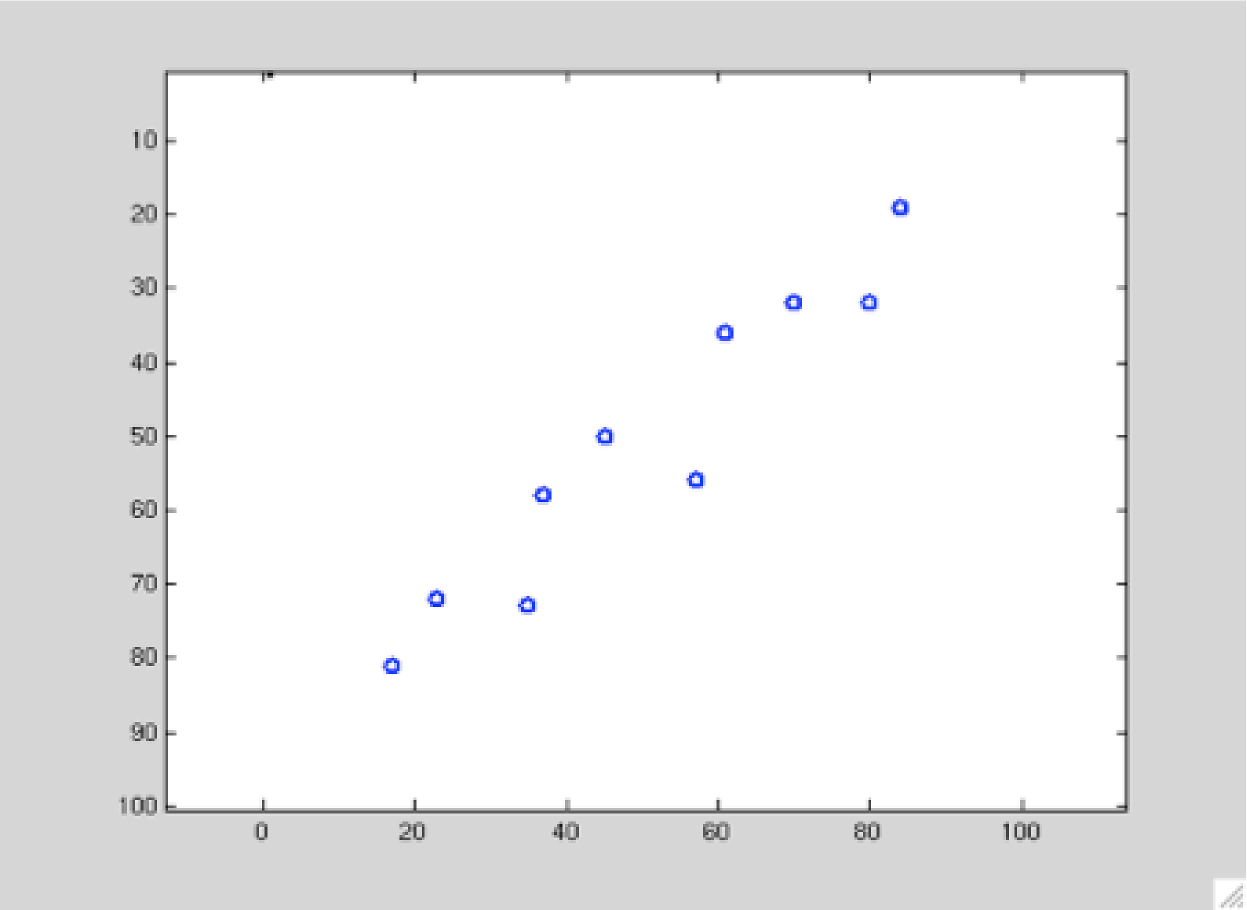




Fitting

- We need to estimate parameters for a model that **optimally** generalize data distribution
- Potential approaches:
  - Least Squares (LS)
  - Robust Statistics
  - RANSAC

LS line fitting



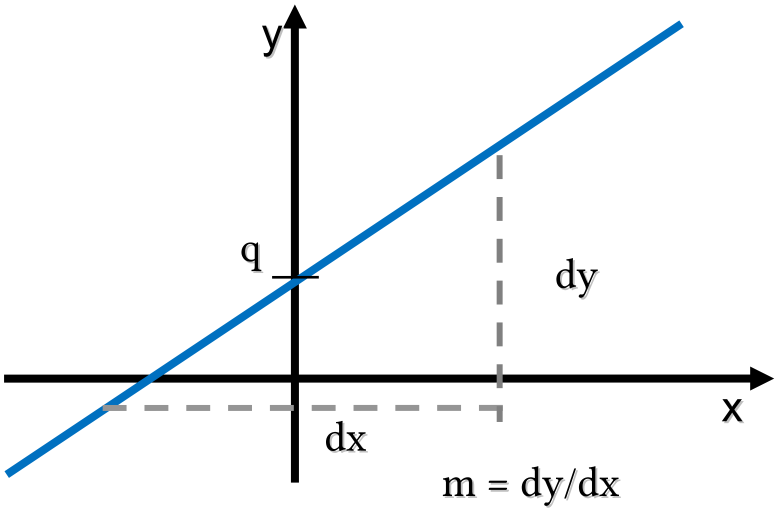
LS line fitting

- Choose a model and its parameters
- Define a error function E($model_i$, data) to evaluate $model_i$
- Adapt model to data
  - find the parameters set that minimize E()

LS Line Fitting

- Model: $y = m\,x + q$
- Parameters: m q

$$\theta = \begin{bmatrix} m \\ q \end{bmatrix}$$



LS line fitting

- Choose a model and its parameters
- Define a error function E($model_i$, data) to evaluate $model_i$
- Adapt model to data
  - find the parameters set that minimize E()

LS Line Fitting

- Error function is E($model_i$, data)
  - The value should indicate how much well $model_i$ fits given data
- For LS Error Function is the sum of **squared residuals**

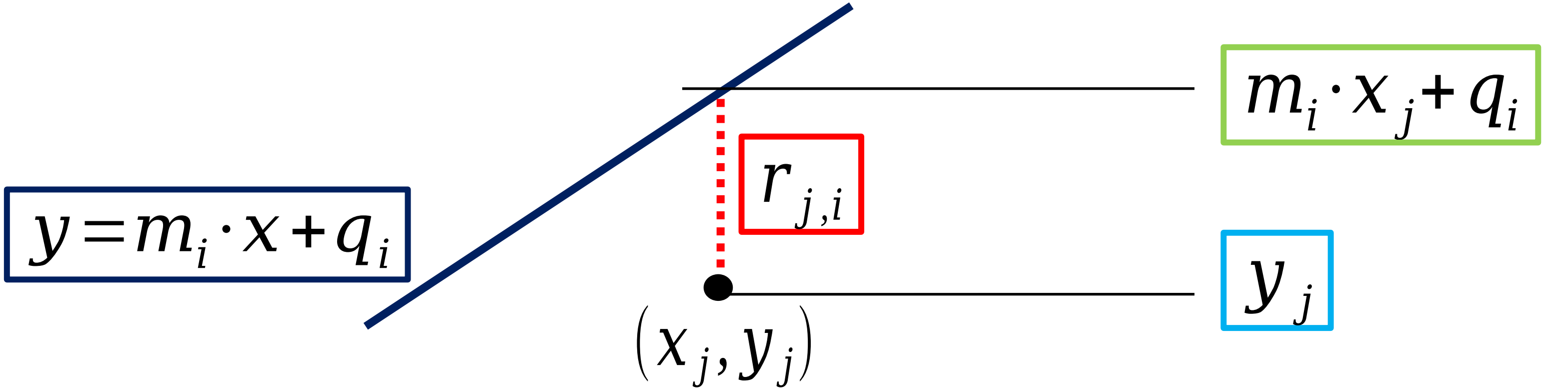
$$E(model_i, data) = \sum_{j=1}^{n} r_{i,j}^2$$

- $r_{i,j}$ is the "difference" between $model_i$ and $data_j$
- We need to define the residual *function*

LS Line Fitting

- For a line fitting we can define residuals as

$$\underbrace{r_{i,j}}_{\text{Distance between data}_j\text{ and model}_i}=\underbrace{y_j}_{\text{Real value of data}_j}-\underbrace{\left(m_i\cdot x_j+q_i\right)}_{\text{Estimated value for data}_j\text{ given by model}_i}$$



LS Line Fitting

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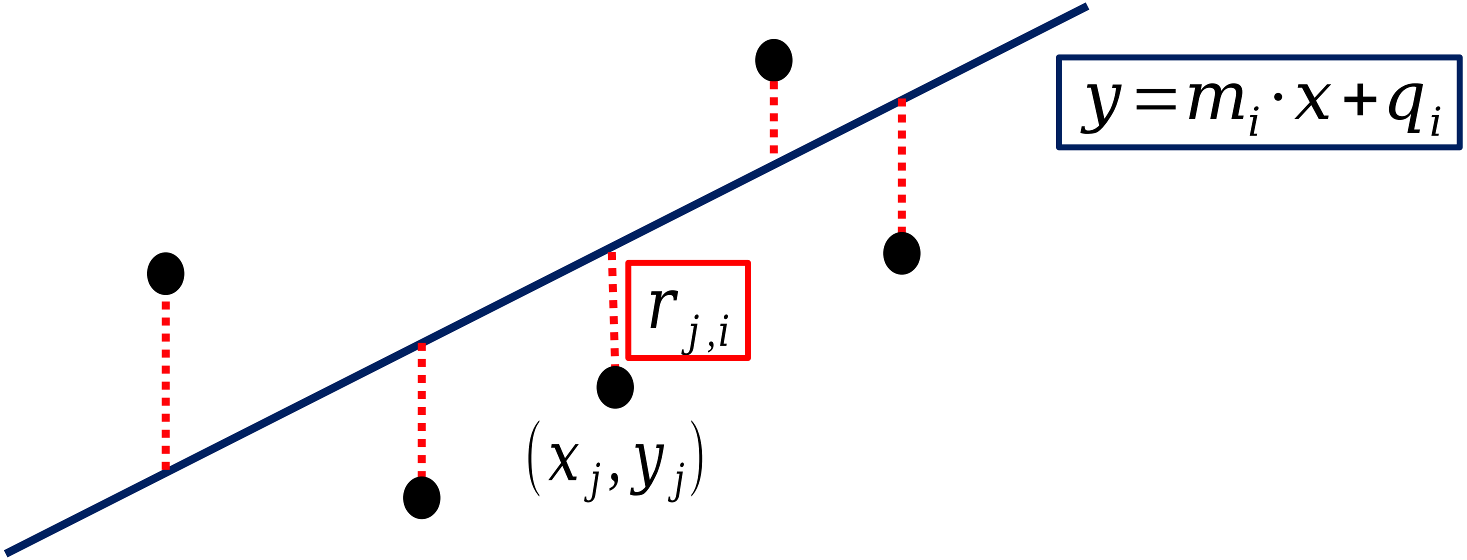
- Given Error Function definition:

$$E\left(model_i, data\right) = \sum_{j=1}^{n} r_{i,j}^2 = \sum_{j=1}^{n} \left[y_j - \left(m_i \cdot x_j + q_i\right)\right]^2$$

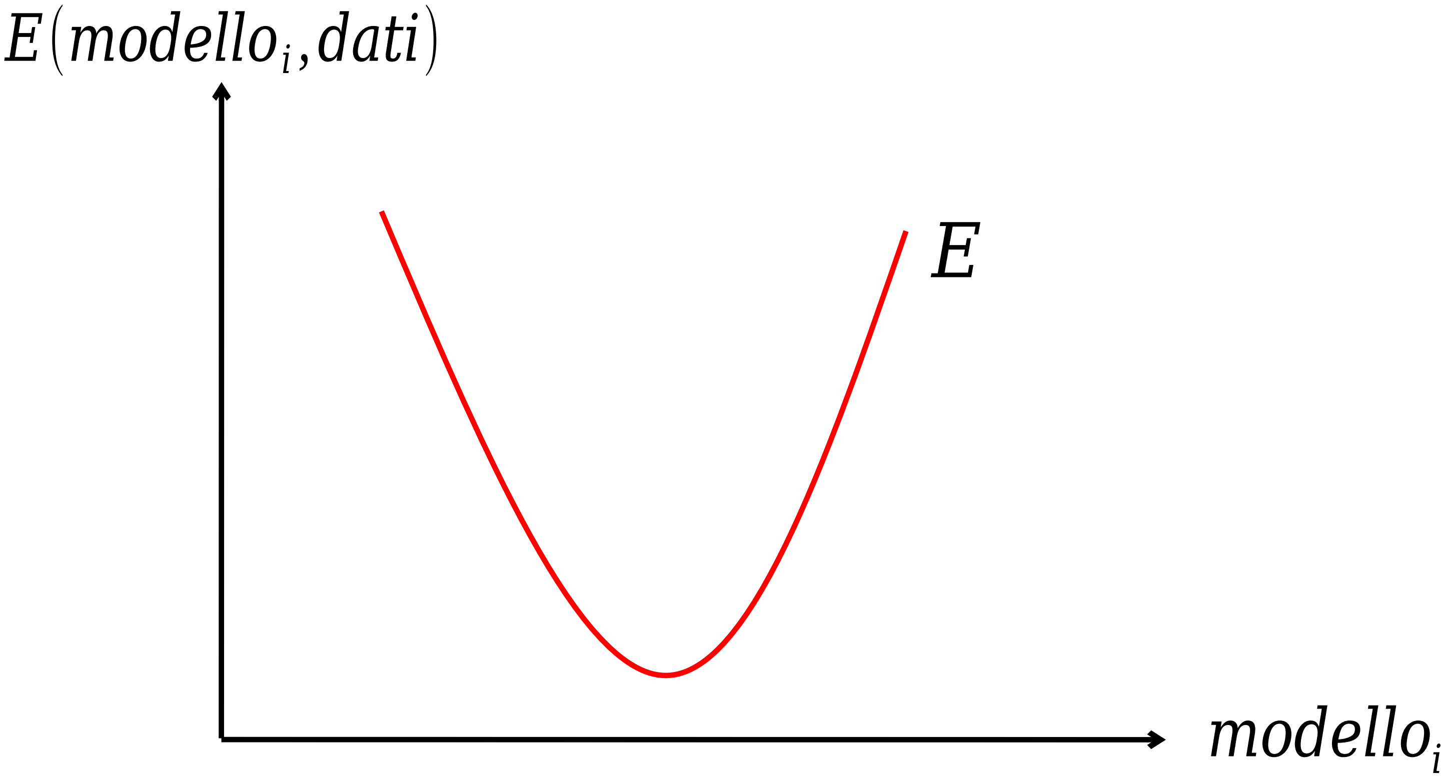


LS line fitting

- Choose a model and its parameters
- Define a error function E($model_i$, data) to evaluate $model_i$
- Adapt model to data
  - find the parameters set that minimize E()

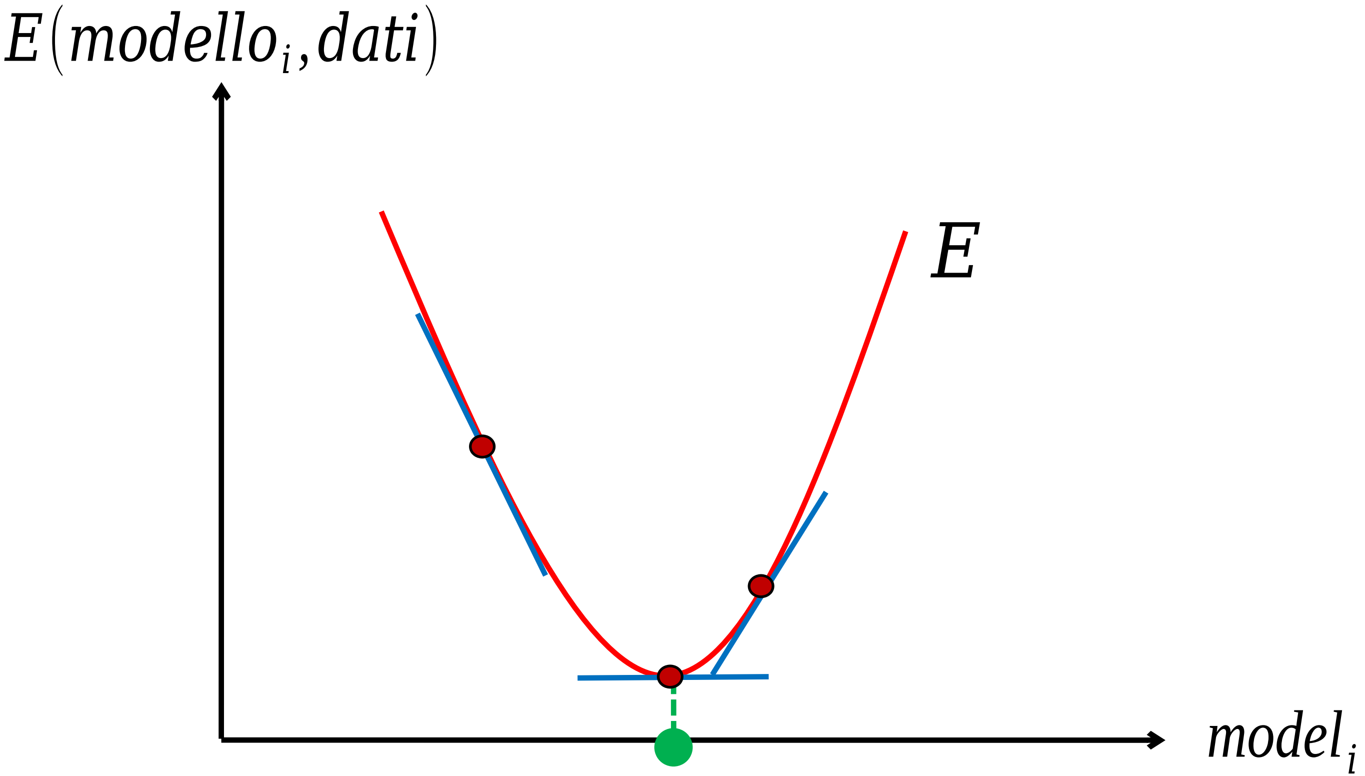
LS Line Fitting

- Consider E() graph



LS Line Fitting

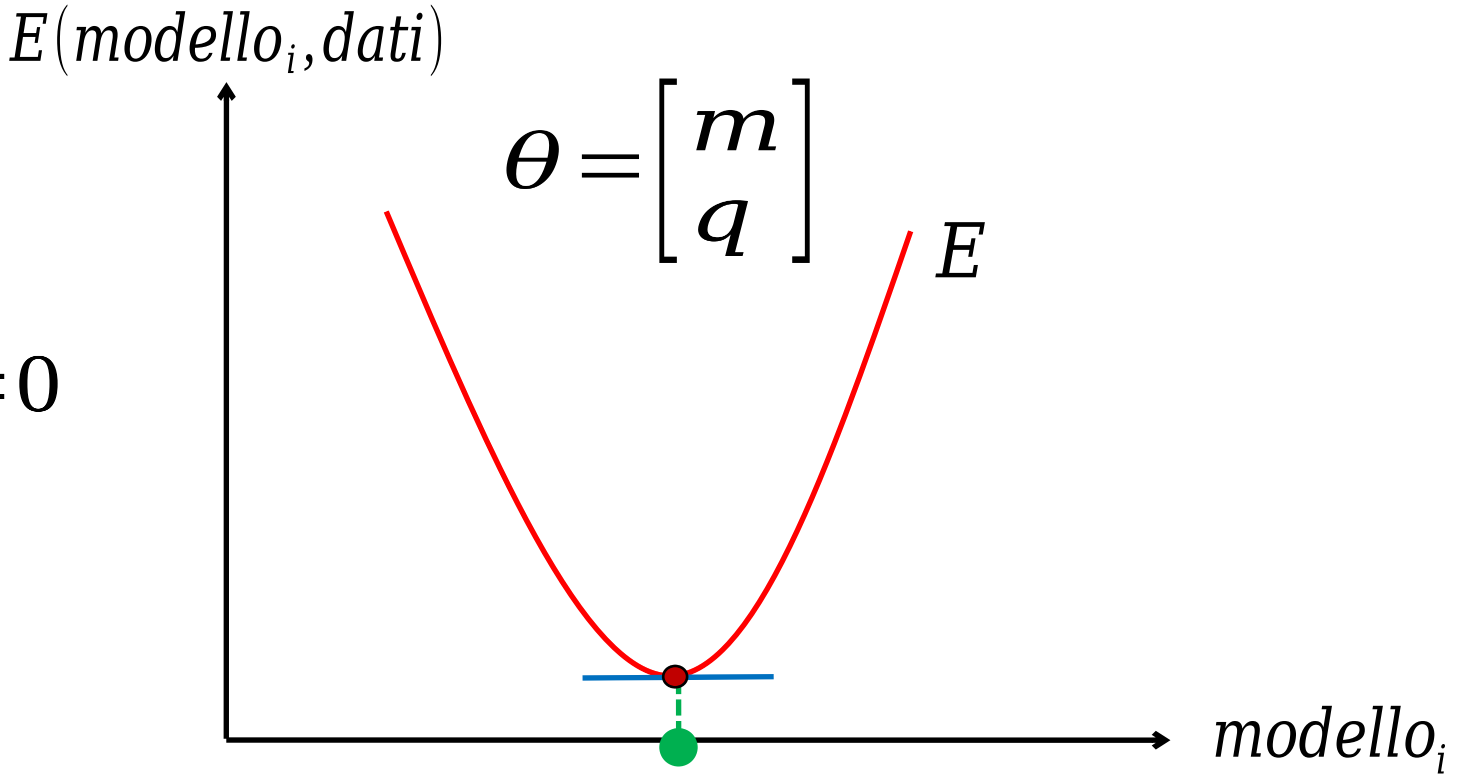
- We need to find the parameter set that minimize E()



LS Line Fitting

- Compute gradient and try to obtain 0

$$\frac{\partial E}{\partial \theta} = \begin{bmatrix} \dfrac{\partial E}{\partial m} \\ \dfrac{\partial E}{\partial q} \end{bmatrix} = 0$$



LS Line Fitting

- Partial derivative of E with respect to m

$$\frac{\partial E}{\partial m}=\frac{\partial}{\partial m}\sum_{j=1}^{n}\left(y_j-\left(m x_j+q\right)\right)^2=0$$

$$.=\frac{\partial}{\partial m}\sum_{j=1}^{n}\left(y_j^2-2y_j\left(x_j m+q\right)+\left(m x_j+q\right)^2\right)$$

$$.=\sum_{j=1}^{n}\frac{\partial}{\partial m}\left(y_j^2-2x_j y_j m-2y_j q+x_j^2 m^2+2x_j q m+q^2\right)$$

$$.=\sum_{j=1}^{n}\left(2x_j^2 m-2x_j y_j+2x_j q\right)$$

LS Line Fitting

- Partial derivative of E with respect to m

$$\frac{\partial E}{\partial m} = \sum_{j=1}^{n} \left(2 x_j^2 m - 2 x_j y_j + 2 x_j q \right) = 0$$

$$\left(\sum_{j=1}^{n} x_j^2 \right) m + \left(\sum_{j=1}^{n} x_j \right) q = \sum_{j=1}^{n} x_j y_j \qquad \text{(Eq. 1)}$$

LS Line Fitting

- Partial derivative of E with respect to q

$$\frac{\partial E}{\partial q}=\frac{\partial}{\partial q}\sum_{j=1}^{n}\left(y_j-\left(m x_j+q\right)\right)^2=0$$

$$.=\frac{\partial}{\partial q}\sum_{j=1}^{n}\left(y_j^2-2y_j\left(x_j m+q\right)+\left(m x_j+q\right)^2\right)$$

$$.=\sum_{j=1}^{n}\frac{\partial}{\partial q}\left(y_j^2-2x_j y_j m-2y_j q+x_j^2 m^2+2x_j q m+q^2\right)$$

$$.=\sum_{j=1}^{n}\left(2x_j^2 m+2q-2y_j\right)$$

LS Line Fitting

- Partial derivative of E with respect to q

$$\frac{\partial E}{\partial q} = \sum_{j=1}^{n} \left(2 x_j m + 2 q - 2 y_j\right) = 0$$

$$\left(\sum_{j=1}^{n} x_j\right) m + \underbrace{\sum_{j=1}^{n} 1}_{n} q = \left(\sum_{j=1}^{n} x_j\right) m + nq = \sum_{j=1}^{n} y_j \quad \text{(Eq. 2)}$$

LS Line Fitting

- We can write equations 1 & 2 in a matrix form

$$\begin{bmatrix} \sum x_j^2 & \sum x_j \\ \sum x_j & n \end{bmatrix} \begin{bmatrix} m \\ q \end{bmatrix} = \begin{bmatrix} \sum x_j y_j \\ \sum y_j \end{bmatrix}$$

LS Line Fitting

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$$X = \begin{bmatrix} x_1 & 1 \\ \vdots & \vdots \\ x_n & 1 \end{bmatrix} \qquad Y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} \qquad \theta = \begin{bmatrix} m \\ q \end{bmatrix}$$

LS Line Fitting

- We can see previous equation as:

$$\underbrace{\begin{bmatrix} \sum x_j^2 & \sum x_j \\ \sum x_j & n \end{bmatrix}}_{X^T X} \underbrace{\begin{bmatrix} m \\ q \end{bmatrix}}_{\theta} = \underbrace{\begin{bmatrix} \sum x_j y_j \\ \sum y_j \end{bmatrix}}_{X^T Y}$$

LS Line Fitting

- We can then solve this as

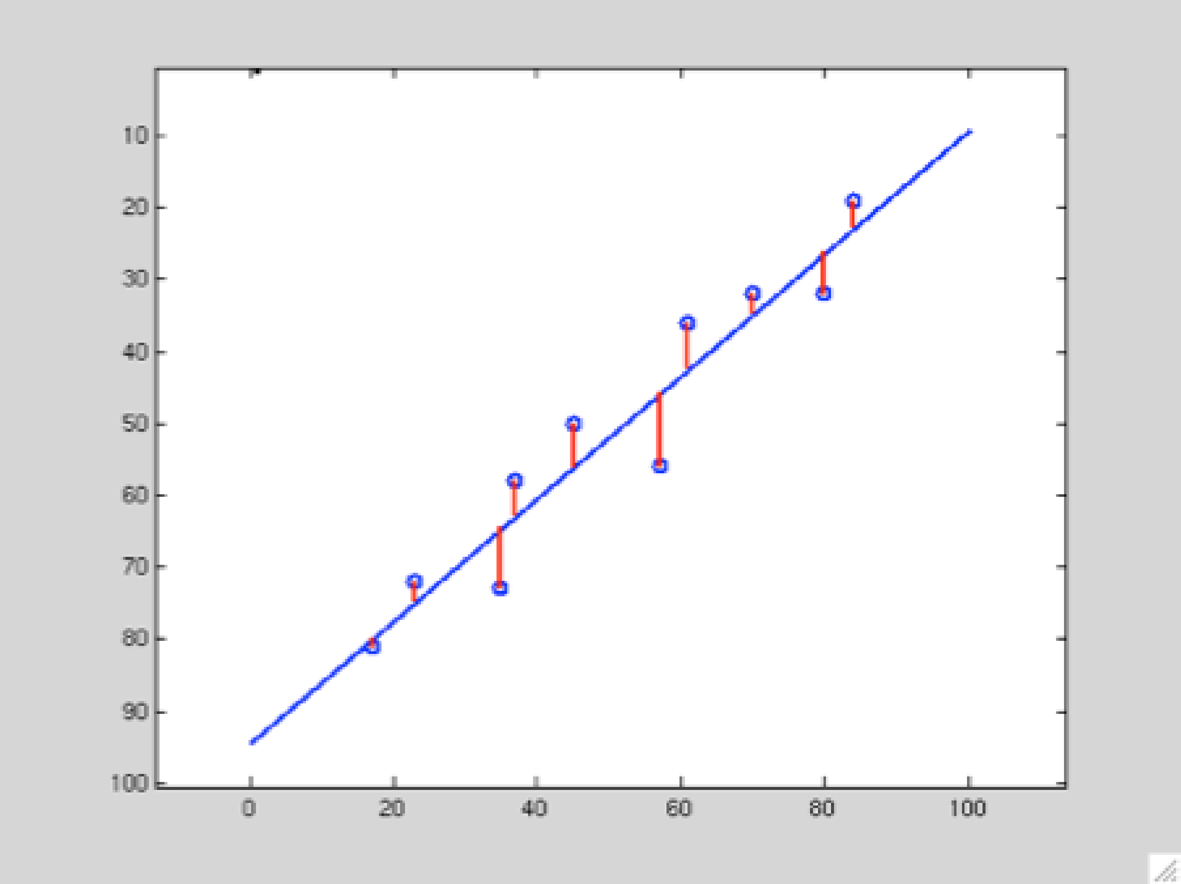
$$X^T X \theta = X^T Y$$

$$\Downarrow$$

$$\theta = \left(X^T X \right)^{-1} X^T Y$$

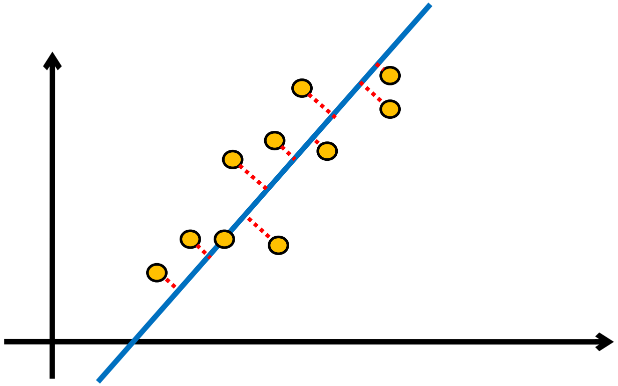
θ is the unknown!

LS Line Fitting



LS Line Fitting – algebraic distance

- (m,q) are not always the best parameter choiche
  - Vertical lines?
- A more general approach is $ax + by + c = 0$
- We can also use algebraic distance as estimator for fitting



LS Line Fitting – algebraic distance

- Cost function E can now be the actual distance

$$E_i = \sum_{j=1}^{n} (a_i x_j + b_i y_j + c_i)^2$$

- Let's find the parameter set that minimize E_i

LS Line Fitting – algebraic distance

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- Partial derivatives can be written as

$$\frac{\partial E}{\partial a} = \sum_{j=1}^{n} \left(2x_j^2 a + 2x_j y_j b + 2x_j c \right)$$

$$\frac{\partial E}{\partial b} = \sum_{j=1}^{n} \left(2x_j y_j a + 2y_j^2 b + 2y_j c \right)$$

$$\frac{\partial E}{\partial c} = \sum_{j=1}^{n} \left(2x_j a + 2y_j b + 2c \right)$$

LS Line Fitting – algebraic distance

- Again, put them as homogeneous equation

$$\underbrace{\begin{bmatrix} \sum x_j^2 & \sum x_j y_j & \sum x_j \\ \sum x_j y_j & \sum y_j^2 & \sum y_j \\ \sum x_j & \sum y_j & n \end{bmatrix}}_{A} \cdot \underbrace{\begin{bmatrix} a \\ b \\ c \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}}_{0}$$

$$A \cdot x = 0$$

LS Line Fitting – algebraic distance

- Then we should solve the homogeneous equation

$$A \cdot x = 0$$

- There is always a solution?
  - No!
- Moreover, ill posed when $A \in R_{3\times 3}$ has a rank<3

LS Line Fitting – algebraic distance

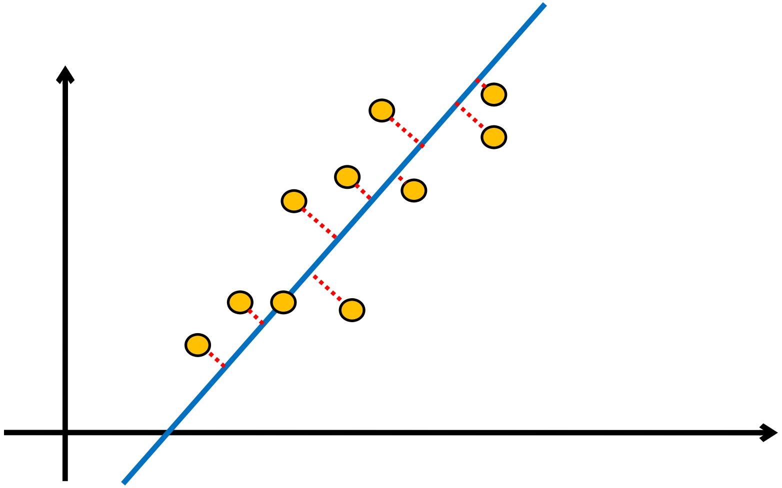
- We can use (again) SVD for a minimization

$$A = UDV^T$$

- x is the last column of V

LS Summary

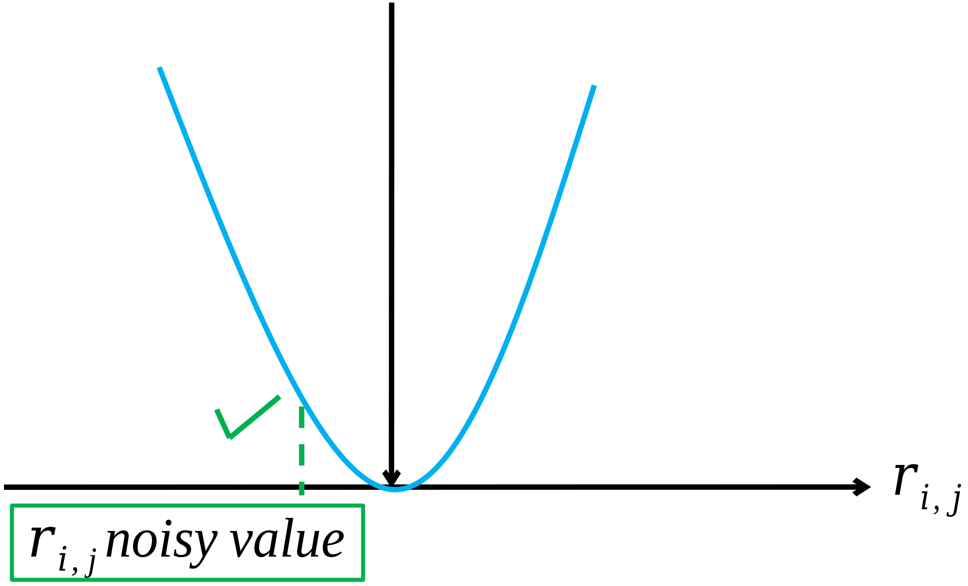
- Summary
  - Simple approach
    - $n \leq 2 \longrightarrow$ closed form
    - $n > 2 \longrightarrow$ SVD
  - No thresholds
  - Results is good when until we have Gaussian noise…



LS issue: outliers

- Not always Gaussian noise
- LS is highly affected by outliers

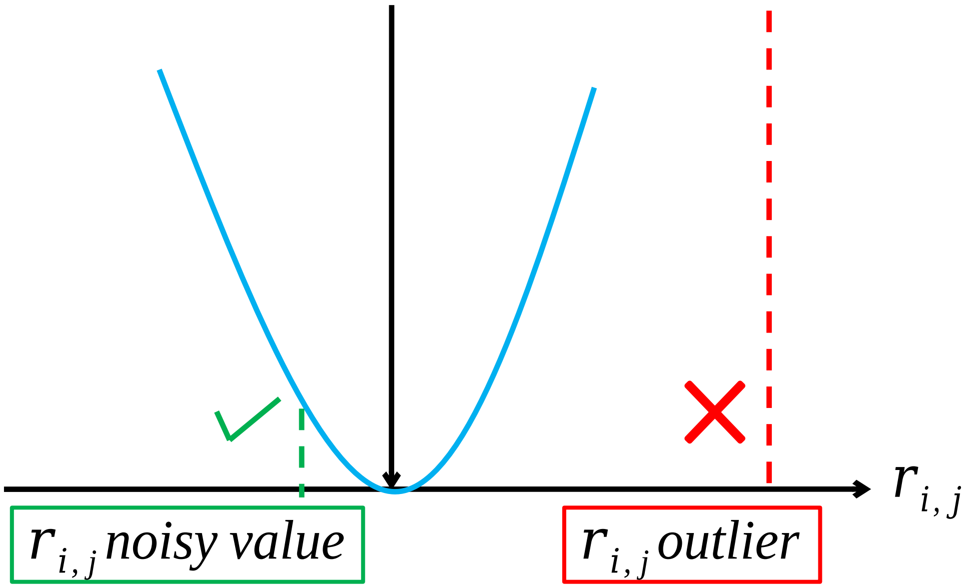
$$f(r_{i,j}) \propto r_{i,j}^2$$



LS issue: outliers

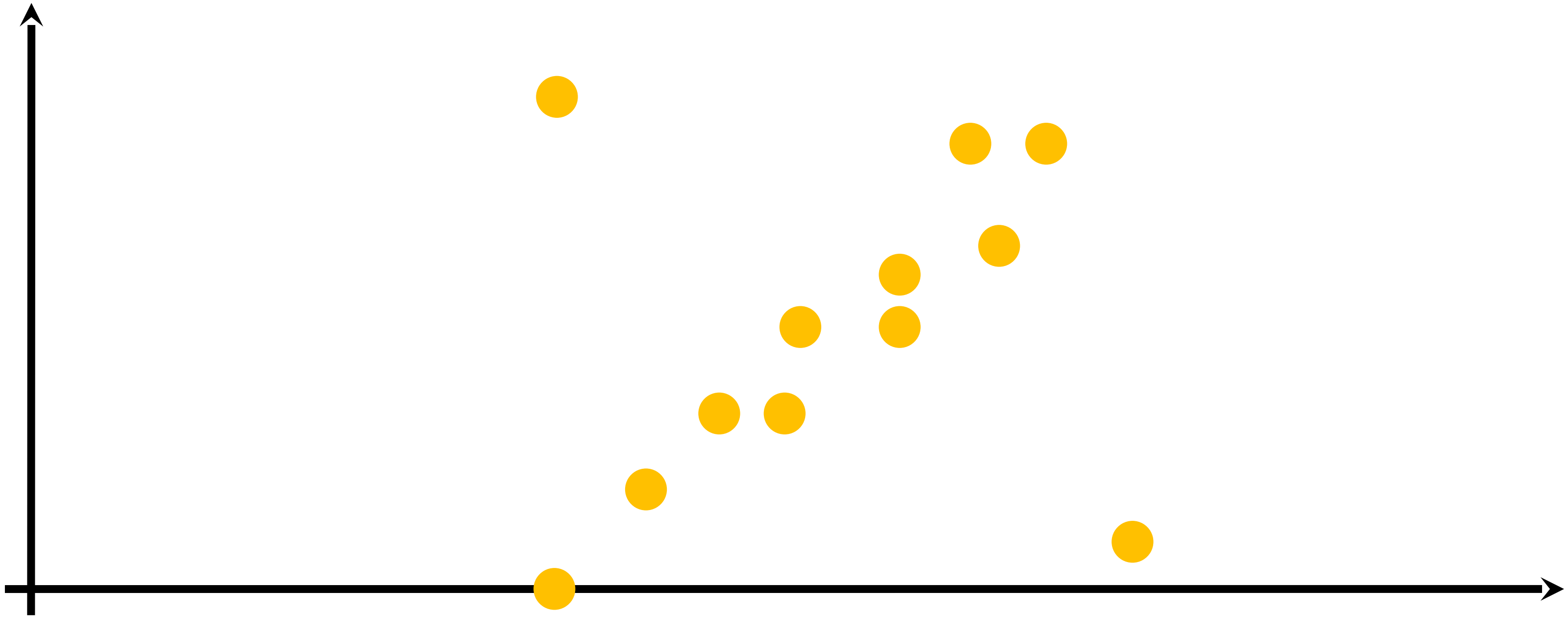
- Not always Gaussian noise
- LS is highly affected by distant points

$$f\left(r_{i,j}\right) \propto r_{i,j}^2$$



LS issue: outliers

- Example: what is the right fitting in this case?



LS issue: outliers

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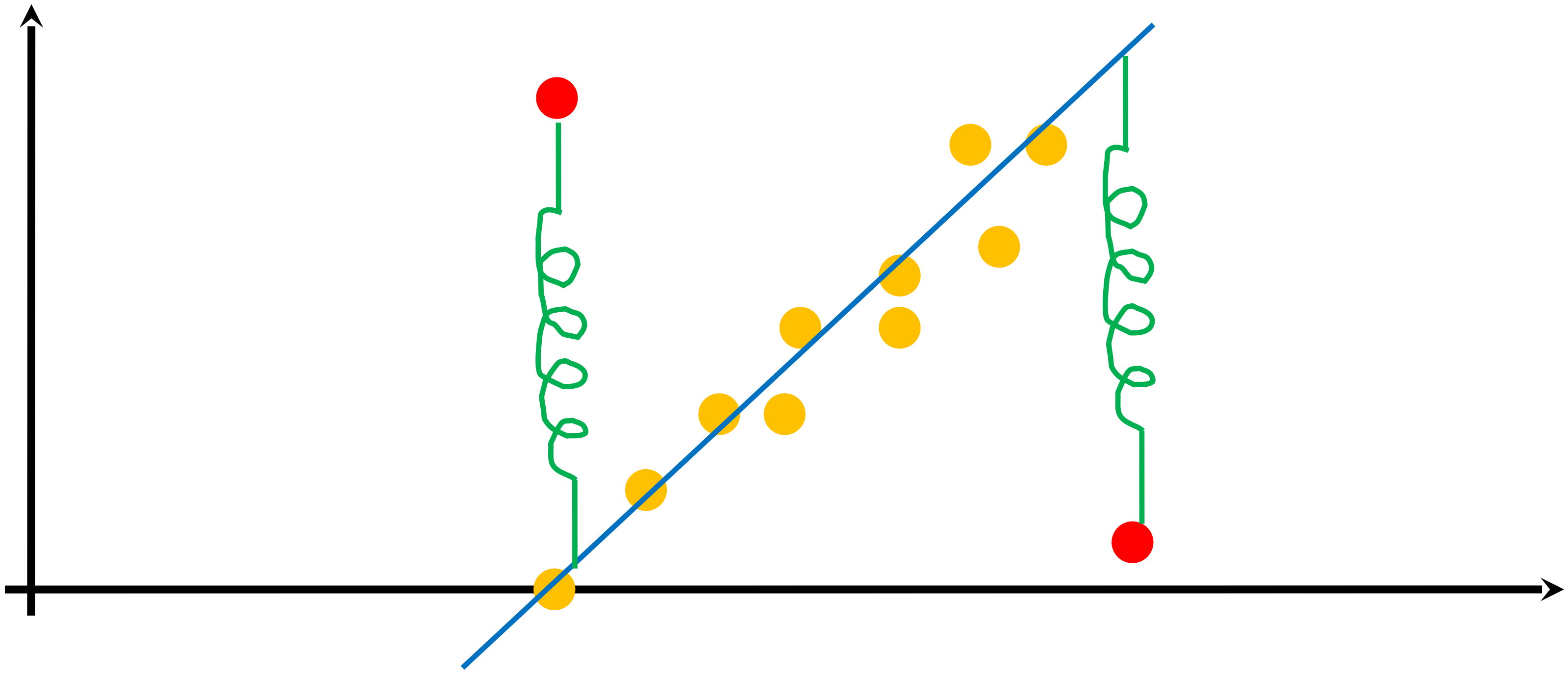

LS issue: outliers

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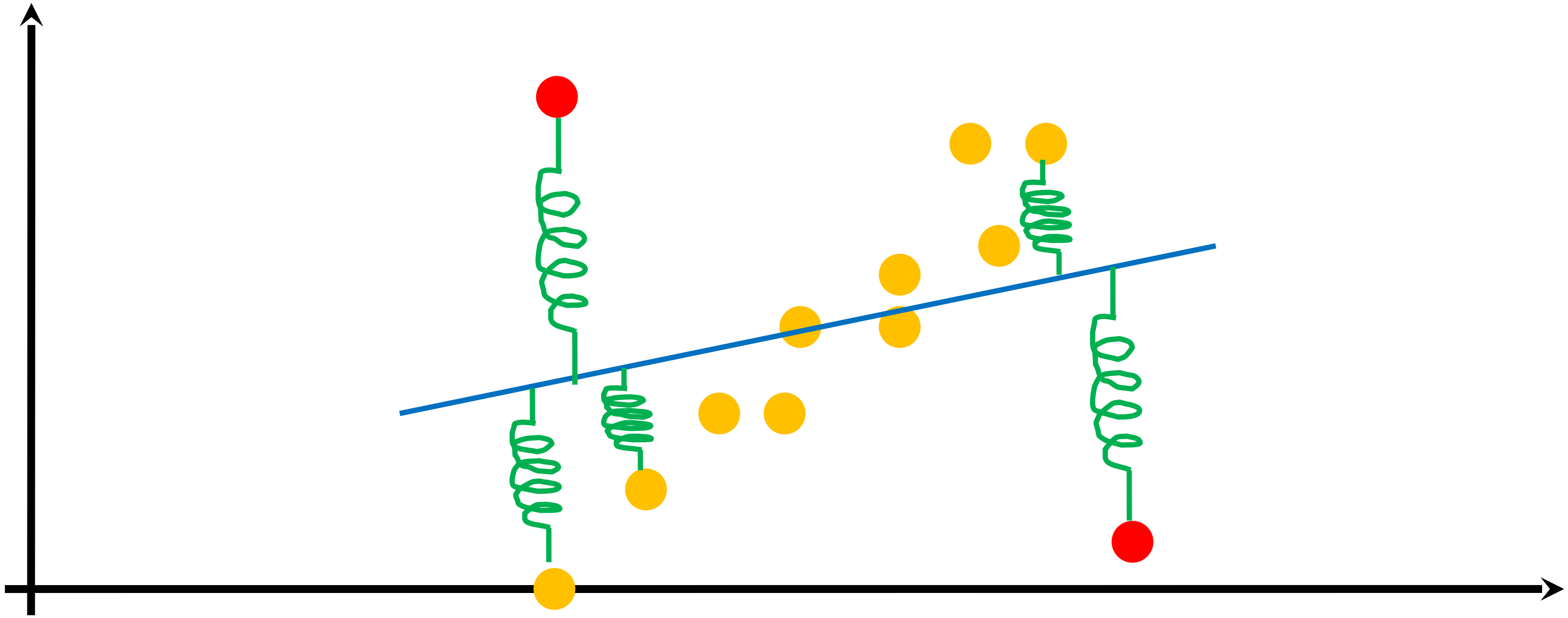
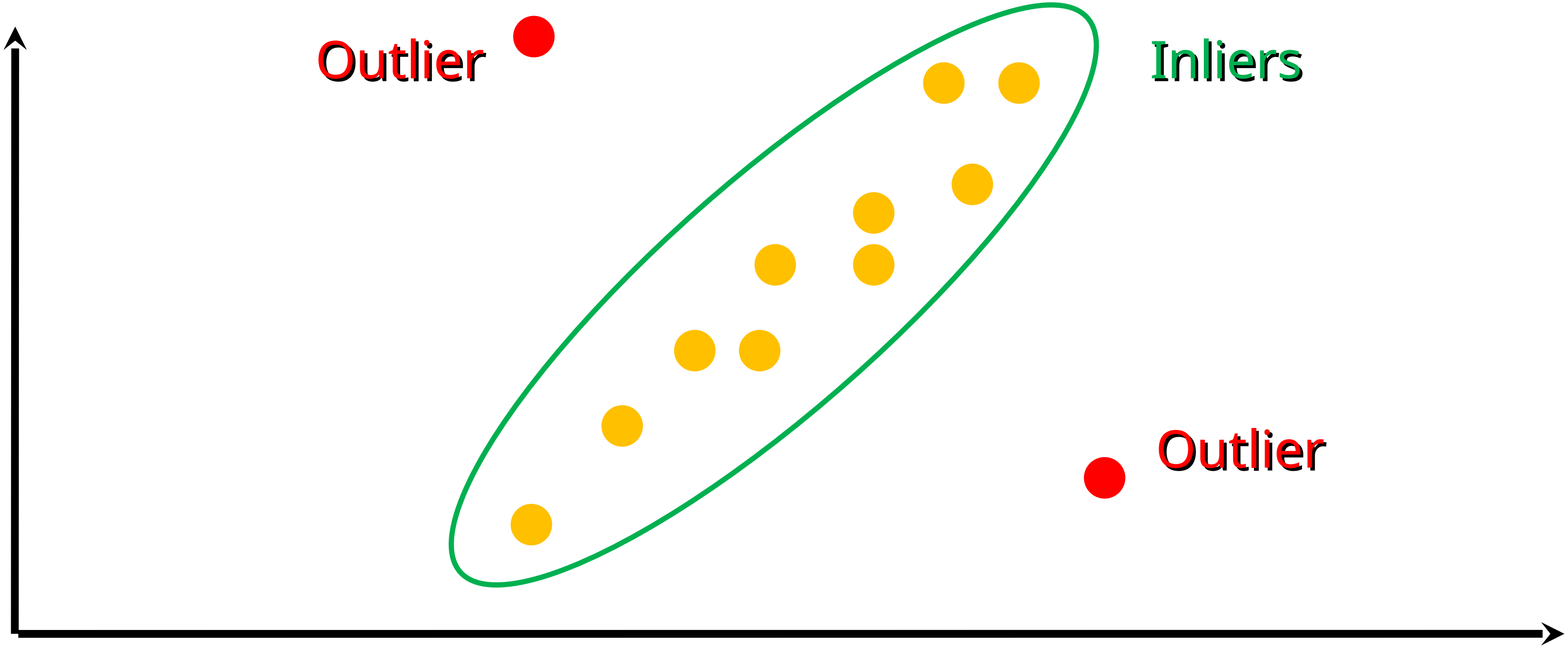


Outliers

- Inliers: data which belong to the model
- Outliers: data not belonging to the model



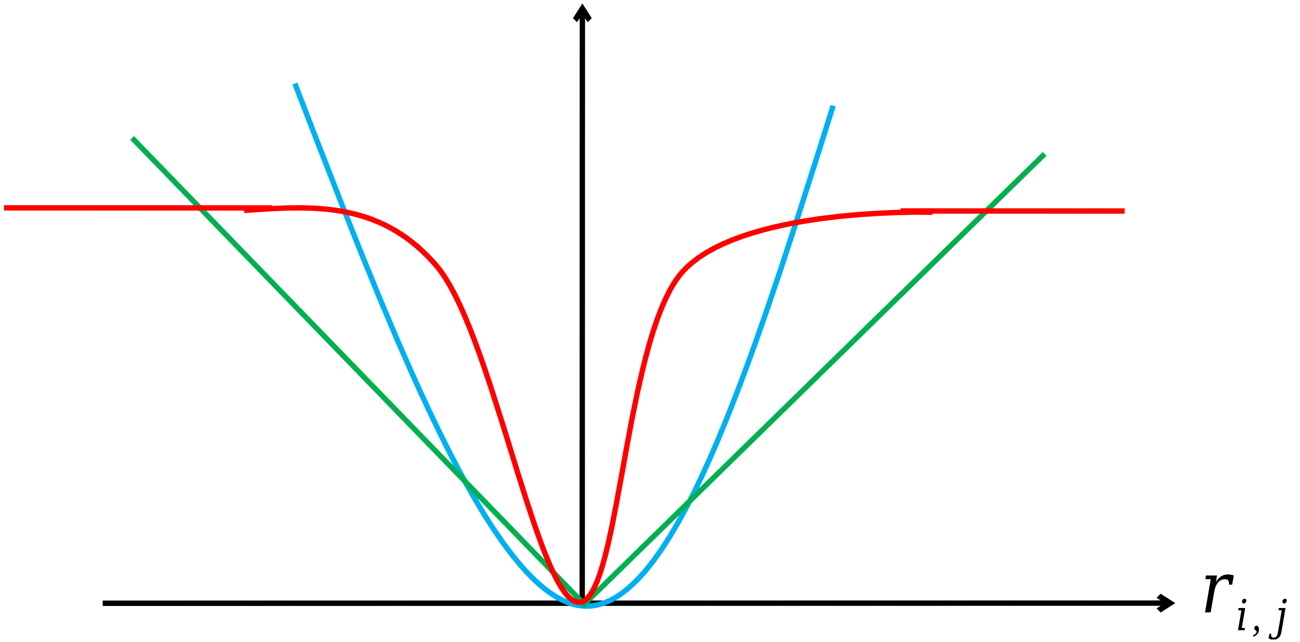
Robust Statistics

- Underlying idea: modify data weight depending on their distance from the model

$$f(r_{i,j}) \propto r_{i,j}^2$$

$$f(r_{i,j}) \propto |r_{i,j}|$$

$$f(r_{i,j}) \propto \frac{r_{i,j}^2}{r_{i,j}^2 + \sigma^2}$$



Robust Statistics

- Instead of squared residuals sum we use a function ρ()

$$\min \sum_{j=1}^{n} r_i^2 \quad \rightarrow \quad \min \sum_{j=1}^{n} \rho(r_i)$$

- The function is symmetrical and features a minimum in zero
  - Loss function

Robust Statistics RANSAC

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- Estimation can also divided in two steps
  - Classify data among Inliers and Outliers
  - Generate a model using inliers only
- RANSAC (RANdom SAmple Consensus) is a widely used approach
  - M. A. Fischler and R. C. Bolles (June 1981). *"Random Sample Consensus: A Paradigm for Model Fitting with Applications to Image Analysis and Automated Cartography"*. Comm. of the ACM 24: 381--395

RANSAC

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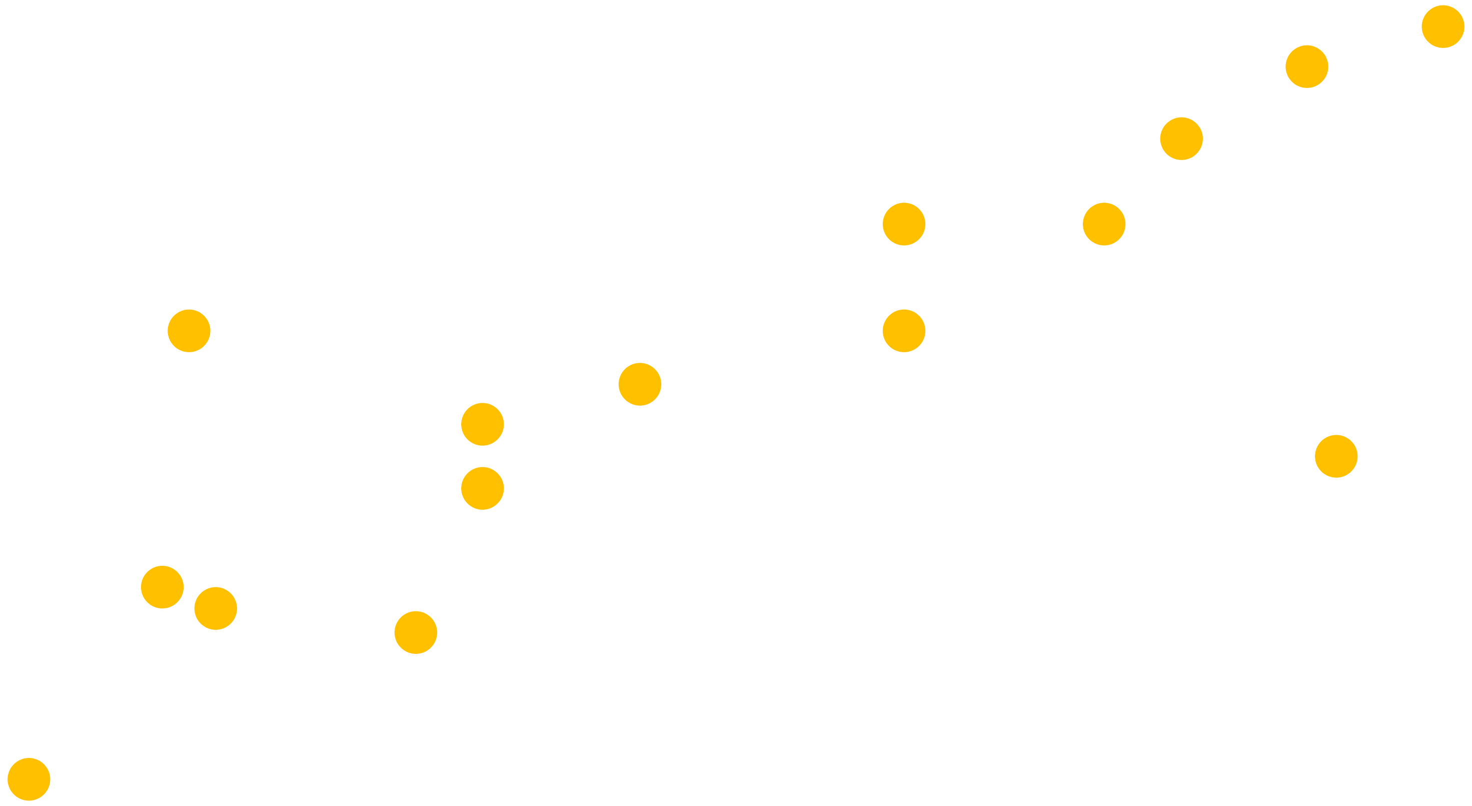
- Given a set of points
  - $(x_1, y_1), (x_2,y_2), \ldots, (x_n,y_n)$
  - Potentially including outliers…
- Estimate parameters
  - Example: $y=mx+q$ or $ax+by+c=0$

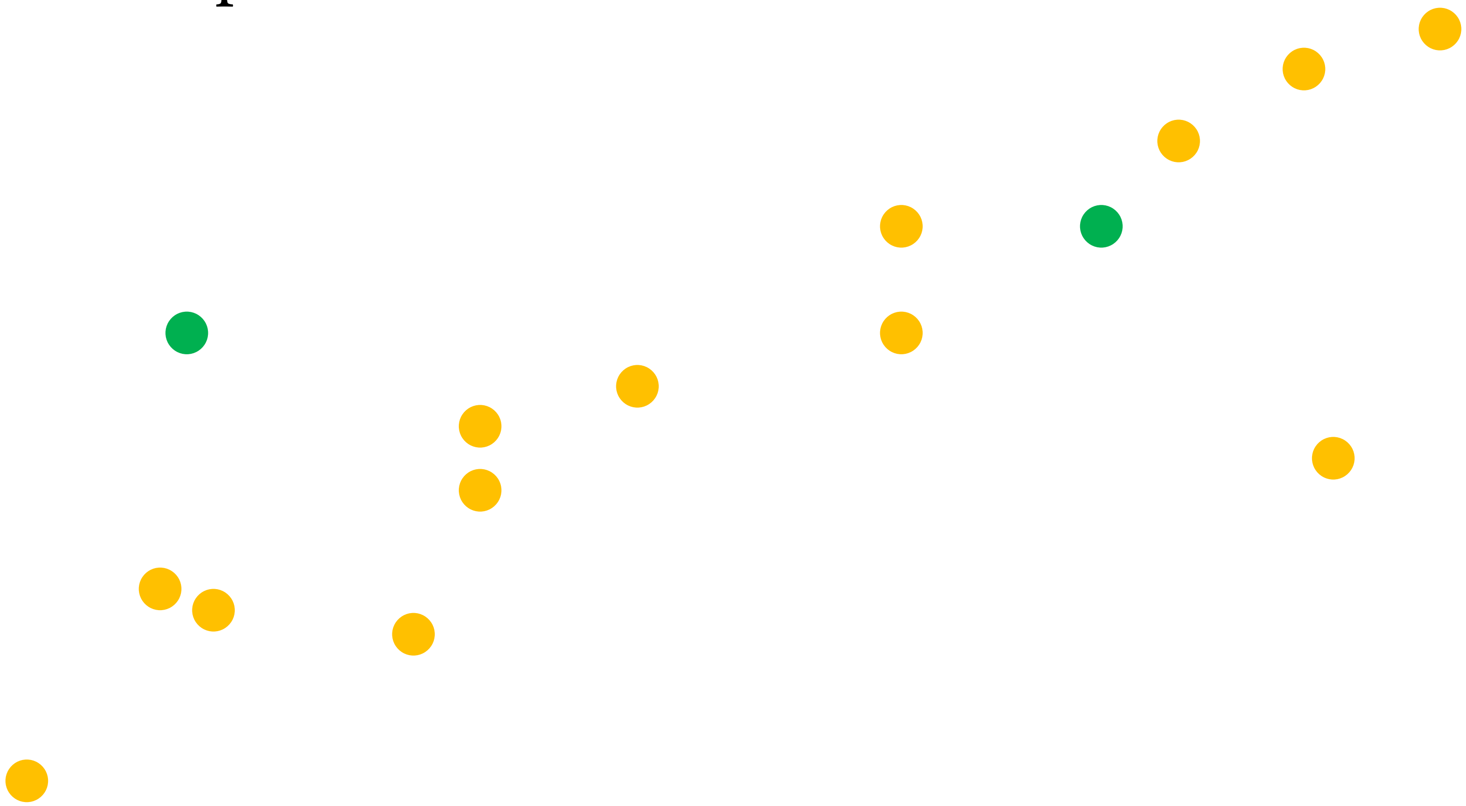
RANSAC line fitting

- We have some outliers…



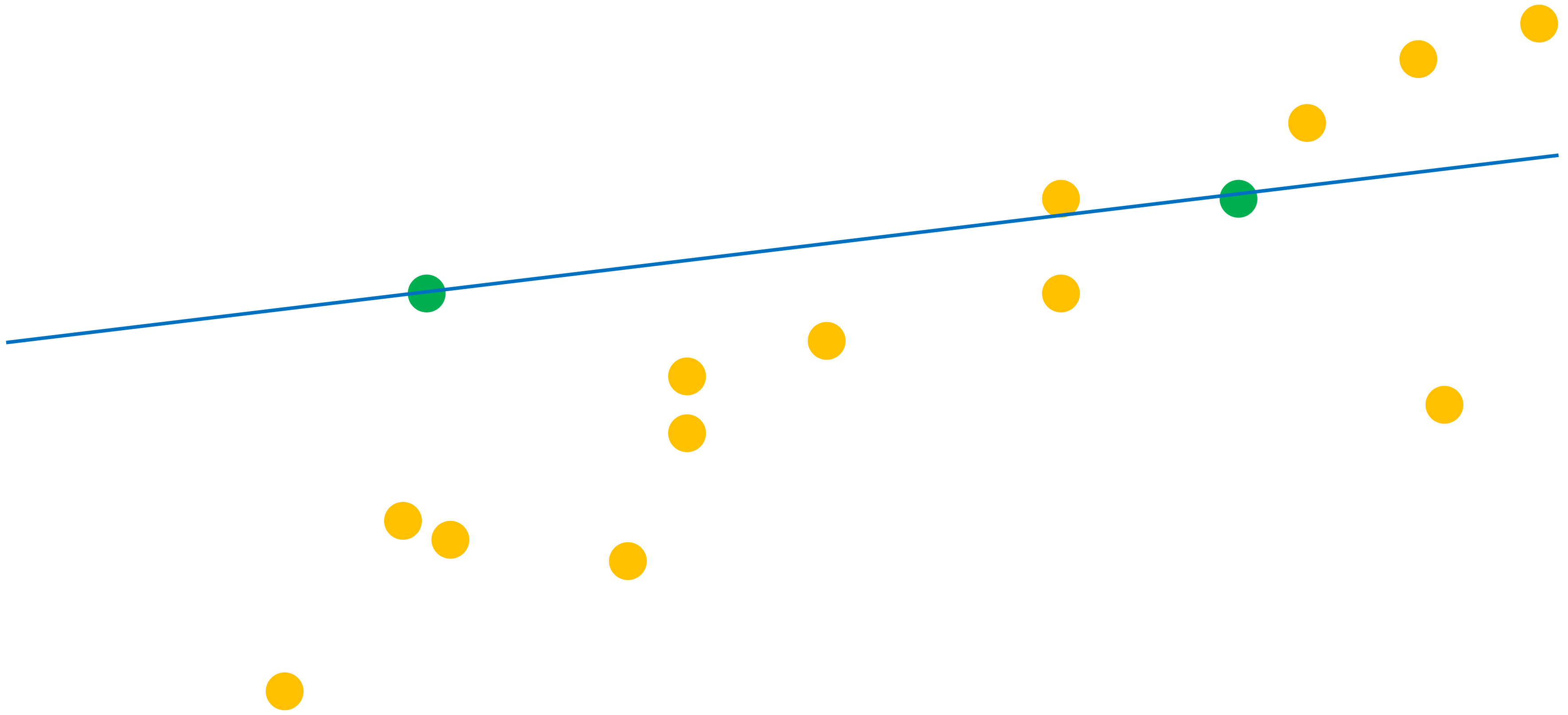
RANSAC line fitting

- Randomly choose 2 points



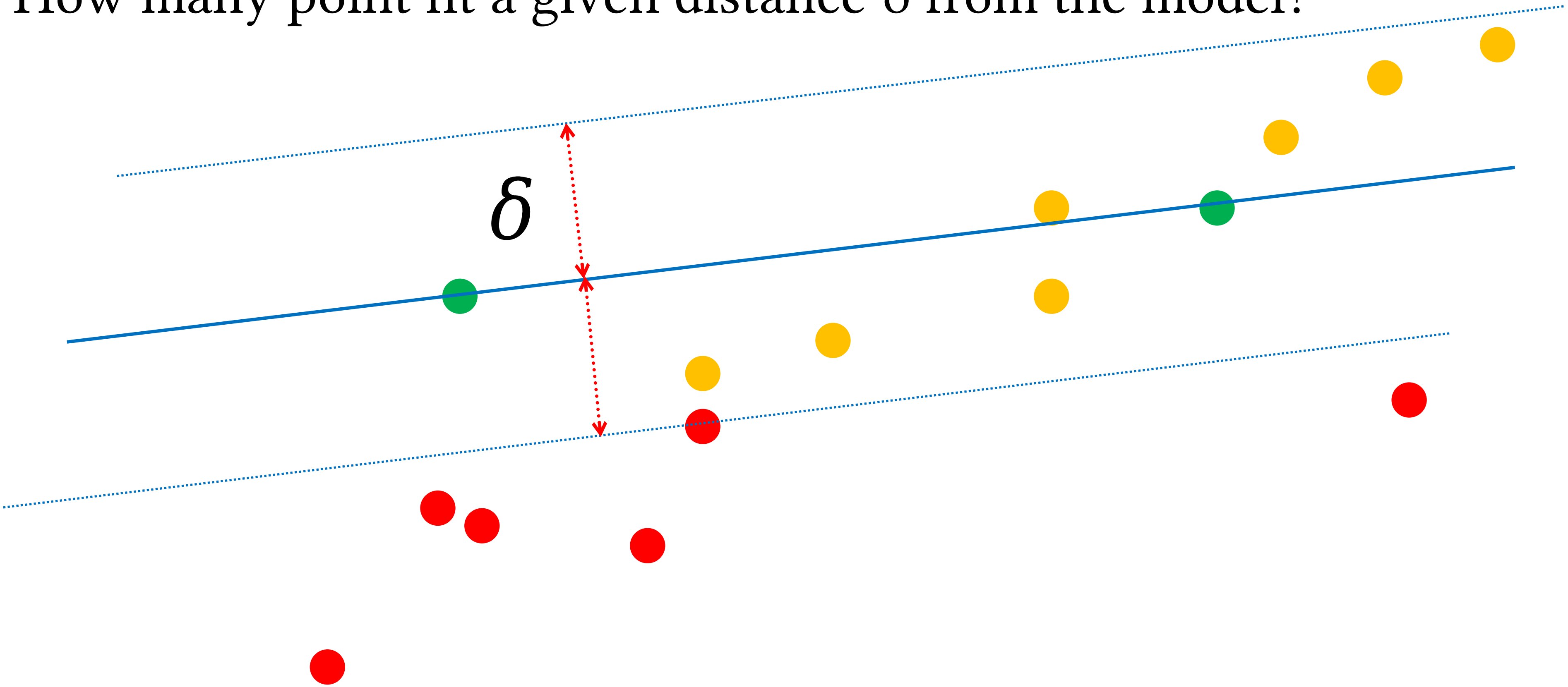
RANSAC line fitting

- Estimate model



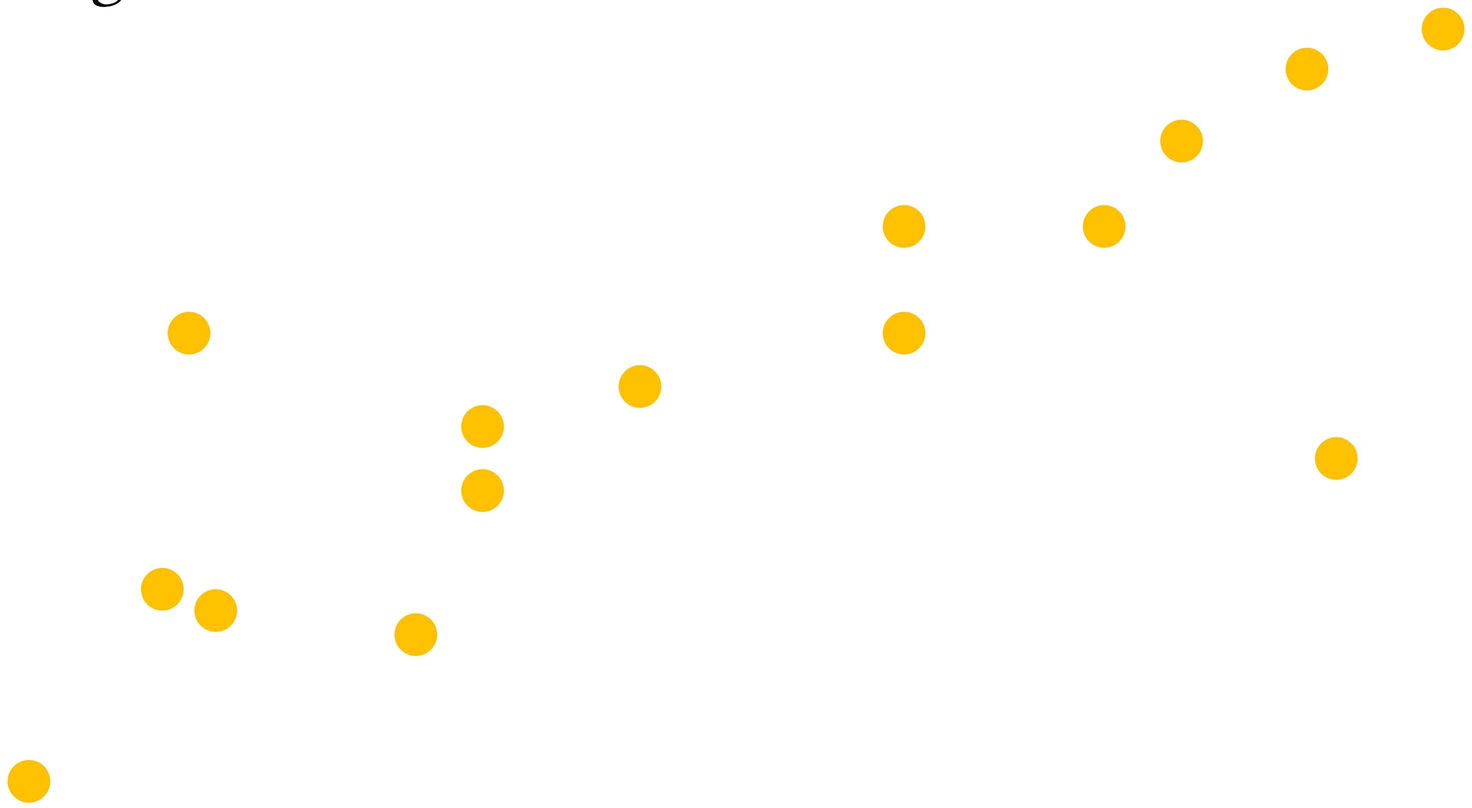
RANSAC line fitting

- How many point fit a given distance δ from the model?



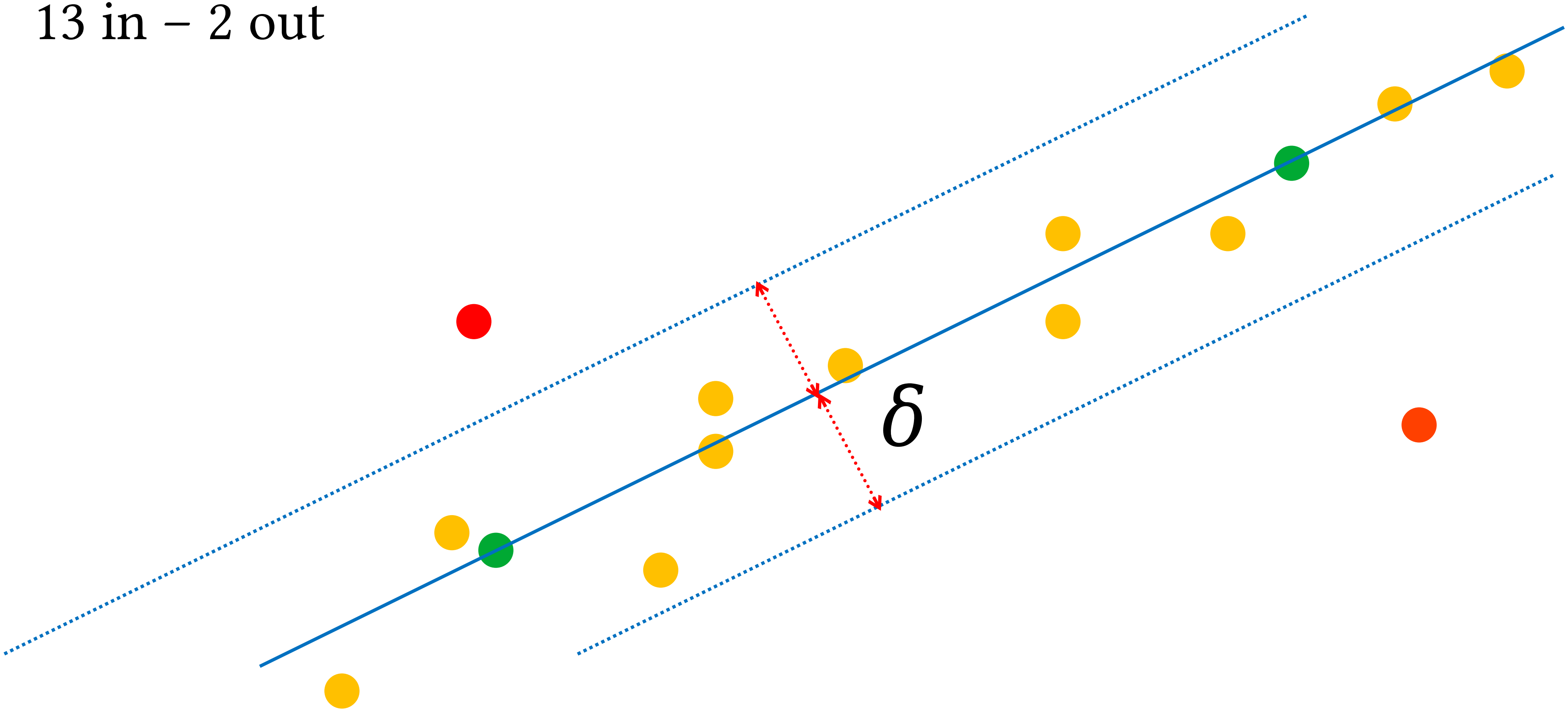
RANSAC line fitting

- Repeat until a "good" result is obtained



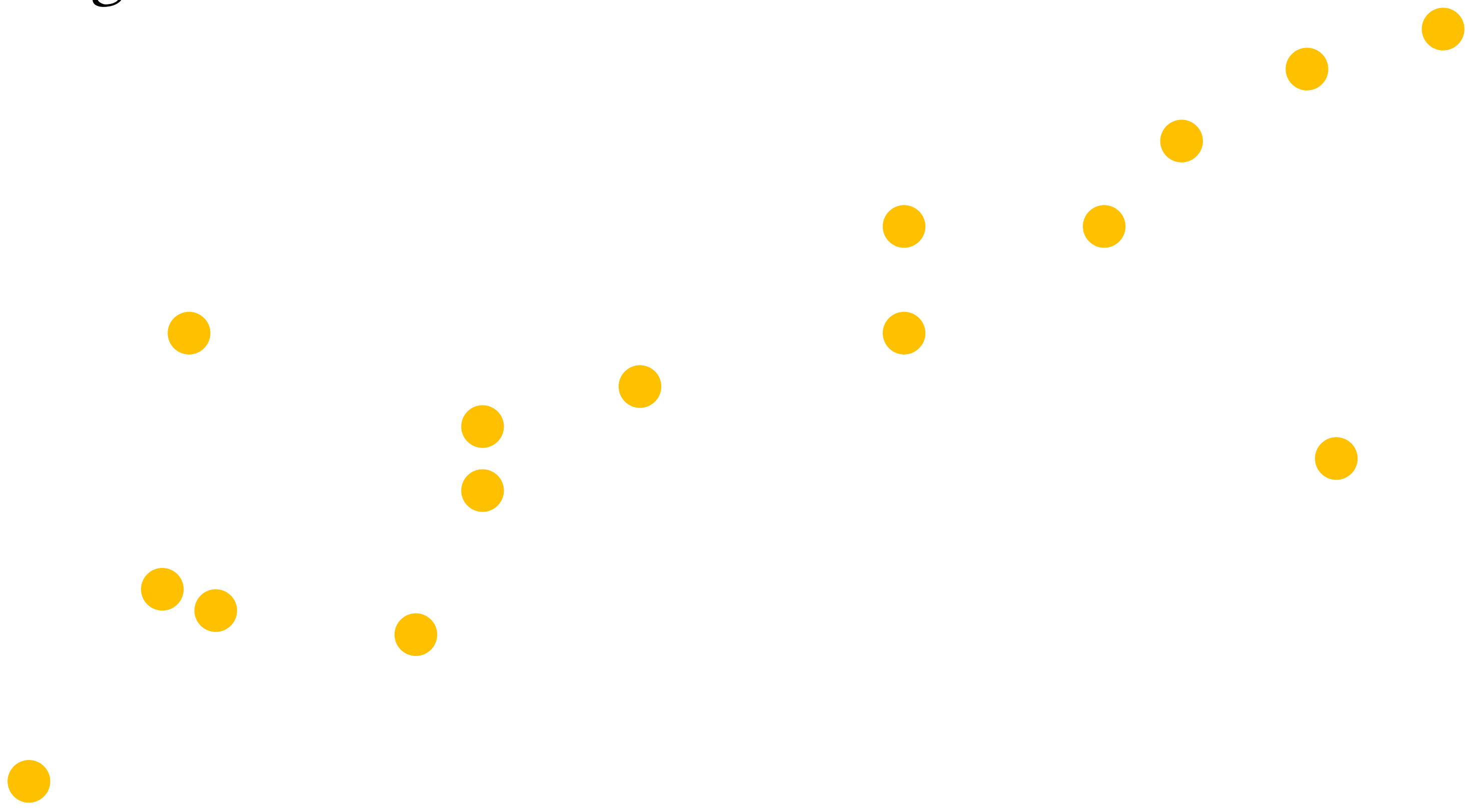
RANSAC line fitting

- 13 in – 2 out



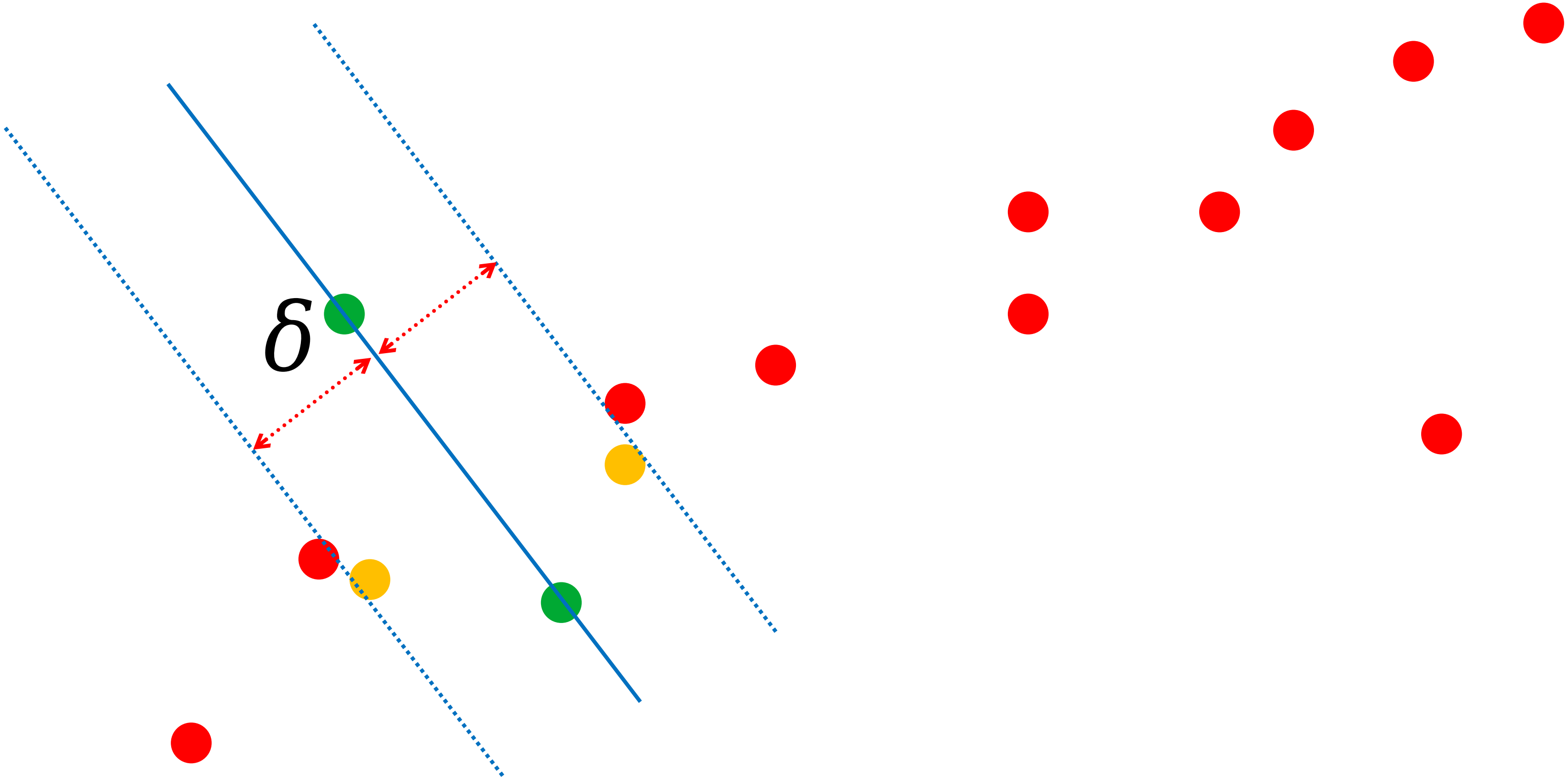
RANSAC line fitting

- Repeat until a "good" result is obtained



RANSAC line fitting

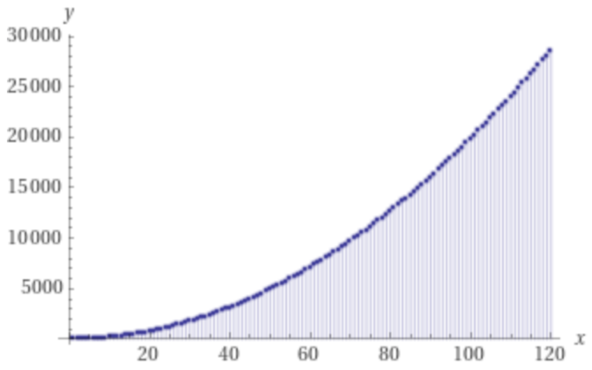
- 4 in…



How many iterations?

- Let's just consider the straight line example
- We have n=15 points
- In order to test a line we need s=2 points

$$\binom{n}{s} = \frac{n!}{(n-s)!\,s!} = 105$$



How many iterations?

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- Given:
  - e $\longrightarrow$ probability a given point is an outlier
  - s $\longrightarrow$ number of points in a subset
  - N $\longrightarrow$ number of subsets (the unknown)
  - p $\longrightarrow$ probability to have at least 1 good subset

$$p=1-\left(1-\left(1-e\right)^{s}\right)^{N}$$

How many iterations?

- Given:
  - e $\longrightarrow$ probability a given point is an outlier
  - s $\longrightarrow$ number of points in a subset
  - N $\longrightarrow$ number of subsets (the unknown)
  - p $\longrightarrow$ probability to have at least 1 good subset

$$p = 1 - \left(1 - \left(1 - e\right)^s\right)^N$$

Inlier Probability

How many iterations?

- Given:
  - e ⟶ probability a given point is an outlier
  - s ⟶ number of points in a subset
  - N ⟶ number of subsets (the unknown)
  - p ⟶ probability to have at least 1 good subset

$$p = 1 - \left(1 - (1 - e)^s\right)^N$$

All Inlier Probability in the Subset

How many iterations?

- Given:
  - e $\longrightarrow$ probability a given point is an outlier
  - s $\longrightarrow$ number of points in a subset
  - N $\longrightarrow$ number of subsets (the unknown)
  - p $\longrightarrow$ probability to have at least 1 good subset

$$p = 1 - \left(1 - (1 - e)^{s} \right)^{N}$$

One or more Outliers Probability in the Subset

How many iterations?

- Given:
  - e $\longrightarrow$ probability a given point is an outlier
  - s $\longrightarrow$ number of points in a subset
  - N $\longrightarrow$ number of subsets (the unknown)
  - p $\longrightarrow$ probability to have at least 1 good subset

$$p = 1 - \left(1 - (1 - e)^s\right)^N$$

Probability that N subsets always contain outliers

How many iterations?

- Given:
  - e $\longrightarrow$ probability a given point is an outlier
  - s $\longrightarrow$ number of points in a subset
  - N $\longrightarrow$ number of subsets (the unknown)
  - p $\longrightarrow$ probability to have at least 1 good subset

$$p = 1 - \left(1 - \left(1 - e\right)^s\right)^N$$

Probability to have at least one uncontaminated subset

How many iterations?

- N can be then computed as f(p,e)
  - s is known given the model

$$\left(1-\left(1-e\right)^{s}\right)^{N}=p-1 \quad \rightarrow \quad N=\frac{\log\left(1-p\right)}{\log\left(1-\left(1-e\right)^{s}\right)}$$

How many iterations?

- Assuming we need a “strong” probability (i.e. p=0.99)

$$N = \frac{\log(1-p)}{\log\left(1-(1-e)^s\right)}$$

	proportion of outliers e						
s	5%	10%	20%	25%	30%	40%	50%
2	2	3	5	6	7	11	17
3	3	4	7	9	11	19	35
4	3	5	9	13	17	34	72
5	4	6	12	17	26	57	146
6	4	7	16	24	37	97	293
7	4	8	20	33	54	163	588
8	5	9	26	44	78	272	1177

How many iterations (N)?

- Again let's consider the straight line example
- We have n=15 points
- We also estimate a ~20% outliers
- In order to test a line we need s=2 points
- We want a p=0.99

$$N = 5 \ll 105$$

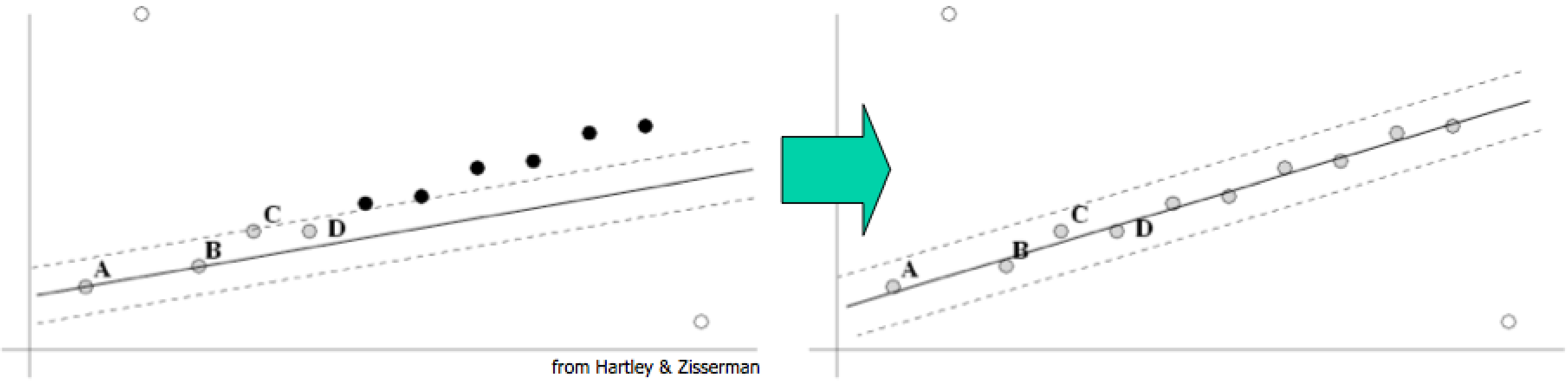
How many iterations?

- Therefore it is not necessary to test all subsets
- We can simply randomly choose N
  - N is usually << all possible combinations
- We can further reduce N
  - Stop iterations when a “sufficient” number of inliers is found

$$T = (1 - e) \cdot n$$

RANSAC: improvements

- Once a model is tested it can be refined considering all inliers
  - LS can be used again



from Hartley & Zisserman

RANSAC: pro&cons

- It often works!
- We need data assumptions
  - inliers/outliers probability
- When several outliers it may be slow...
  - No upper time
  - You can limit iterations $\longrightarrow$ far from optimal solution

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Model Fitting

Question time!

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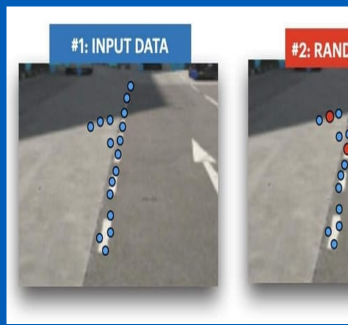
Model Fitting



Model Fitting

Summary

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- What is Model Fitting?
- Is it difficult?
- Least Squares (LS)
  - Line fitting example
  - LS issues (outliers)
- Robust Statistics
- RANSAC

A parte introdurre cosa sia il model fitting ci baseremo su tre tecniche tutte prendendo ad esempio il fitting di una linea (ma ovviamente sono tutte tecniche estendibili)

Model Fitting

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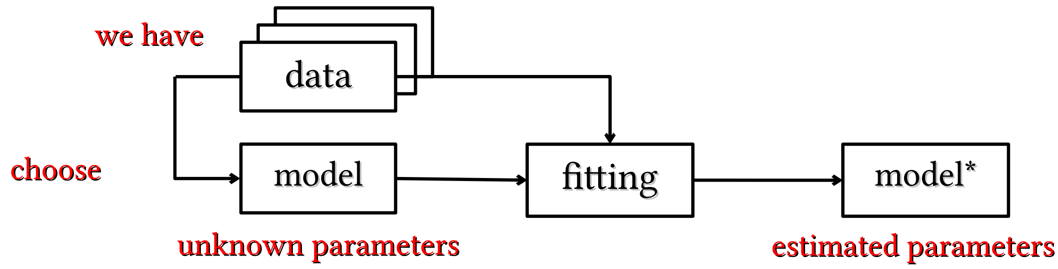
- We have data
  - A lot of them
- Is there a model that generalize data distribution?
  - Choose a model
  - Compute model parameters

Non necessariamente visione
Ma comunemente usato.
Problema ho dei dati (pixel, point cloud, ecc.) e faccio l'ipotesi di conoscenza a priori di un modello. Come fitto il modello?

Model Fitting



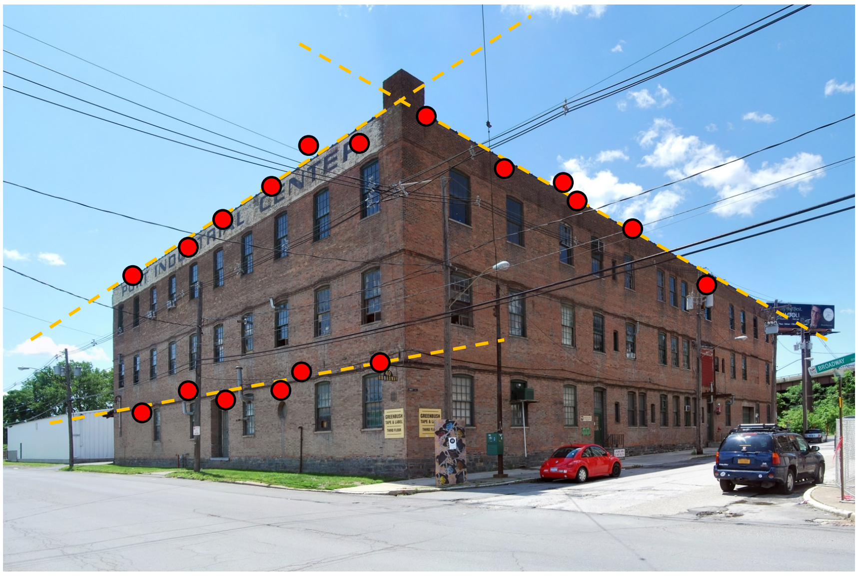
Il modello lo scegliamo noi in base ai dati e a cosa vogliamo estrarre. Però dato il modello voglio poi conoscere i suoi parametri o meglio quelli che descrivono meglio i dati che uso

Models

- Lines, curves, planes…
- Homographies, Calibration Matrices…
- …

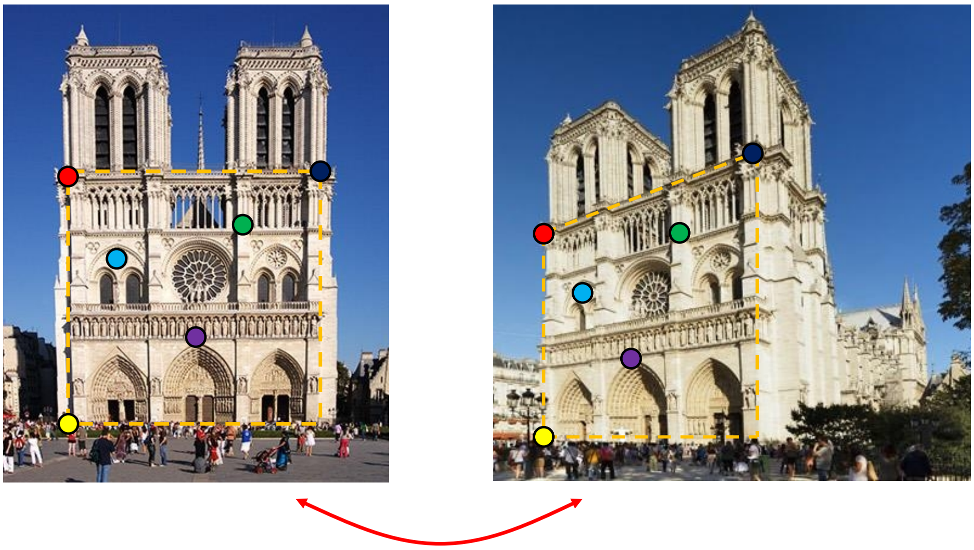
Modelli possono essere il fitting di curve (linee, parabole, piani ecc.) o, e questo lo vedremo, omografie, calibrazioni ecc.

Example: lines fitting



Supponiamo di avere come dati i punti in esame. E so (il modello) che mi descrivono delle linee. Potrei usare hough. Ma i dati sono rumorosi, funzionerebbe? In questo caso probabilmente sì. Ma qui c'è un rumore relativamente piccolo.

Example: homograpy



Date due immagini (stereo ma un po' extreme) e relative corrispondenze.

Supponiamo di voler ricavare la relativa omografica che mi porta il piano di quella facciata sull'altra. Anche qui c'è rumore. Come faccio?

Problems

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- Noise
- Outliers
- Missing data

Non sempre il rumore è la cosa piú critica. Il problema principale sono spesso gli outlier.

Outliers

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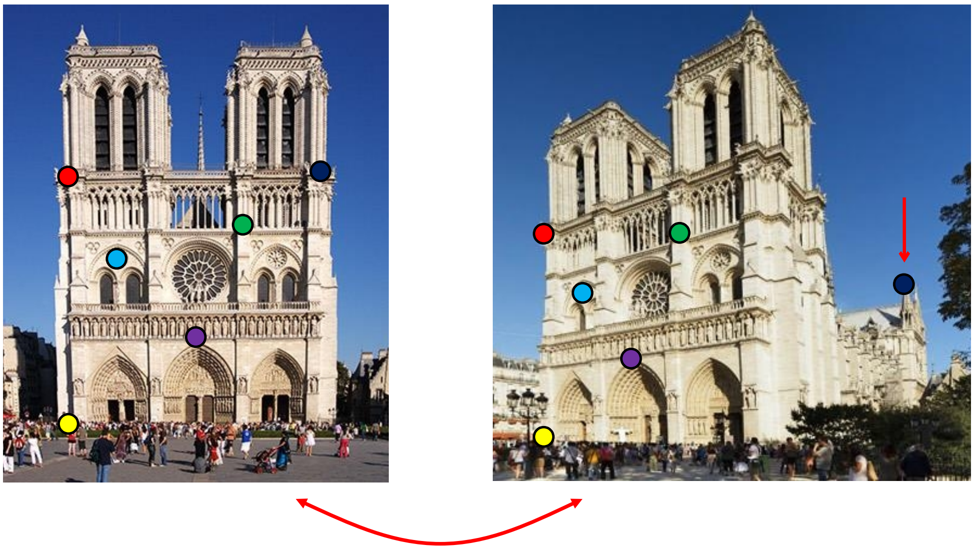

Stessa immagine di prima. Qui però non solo c'è rumore (ovvero un problema di precisione) dei punti selezionati. Ma c'è una coppia di corrispondenze clamorosamente sbagliata. Temo che l'omografia che trovo potrebbe essere clamorosamente sbagliata...

Fitting

- We need to estimate parameters for a model that **optimally** generalize data distribution
- Potential approaches:
  - Least Squares (LS)
  - Robust Statistics
  - RANSAC

Vediamo qualche possibile approccio o meglio di fatto due: minimi quadrati e RANSAC

LS line fitting

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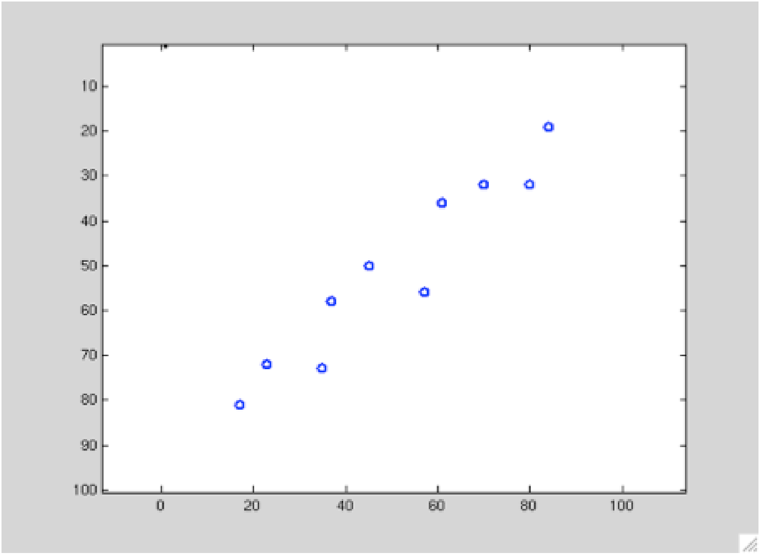

Per semplicità di modello limitiamoci ad un bel line fitting.

LS line fitting

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- Choose a model and its parameters
- Define a error function E($model_i$, data) to evaluate $model_i$
- Adapt model to data
  - find the parameters set that minimize E()

I passi per fare model fitting sono comunque comuni a tutti gli approcci:
- scelgo un modello (di solito è la parte semplice)
- definisco una funzione di errore per valutare un n-upla di parametri del modello
- cerco la n-upla che minimizza E()

LS Line Fitting

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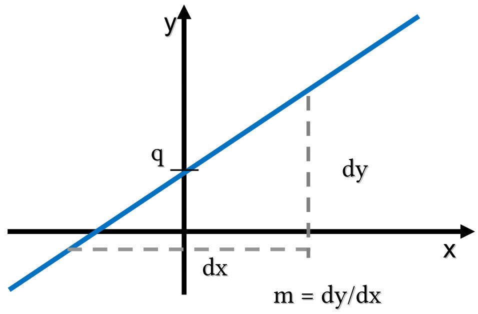
- Model: $y = m\,x + q$
- Parameters: m q

$$\theta = \begin{bmatrix} m \\ q \end{bmatrix}$$



Supponendo di fare line fitting coi minimi quadrati usiamo come modello la classica equazione della retta.
I parametri che voglio trovare sono quindi m e q

LS line fitting

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- Choose a model and its parameters
- Define a error function E(model$_i$, data) to evaluate model$_i$
- Adapt model to data
  - find the parameters set that minimize E()

Definizmo funzione di errore che confronta i parametri o meglio una possibile confiogurazione dei parametri con i dati a disposizione

LS Line Fitting

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- Error function is $E(model_i, data)$
  - The value should indicate how much well $model_i$ fits given data
- For LS Error Function is the sum of **squared residuals**

$$E(model_i, data) = \sum_{j=1}^{n} r_{i,j}^2$$

- $r_{i,j}$ is the “difference” between $model_i$ and $data_j$
- We need to define the residual *function*

Least Squares ⟶ minimi quadrati

Posso pensare come modello la somma dei minimi quadrati della differenza tra modello e dato j-esimo

Quindi se il modello iesimo restituisce 75 e il modello kesimo 27 allora il modello k-esimo è meglio

Sì, ma cosa scelgo come r?

LS Line Fitting

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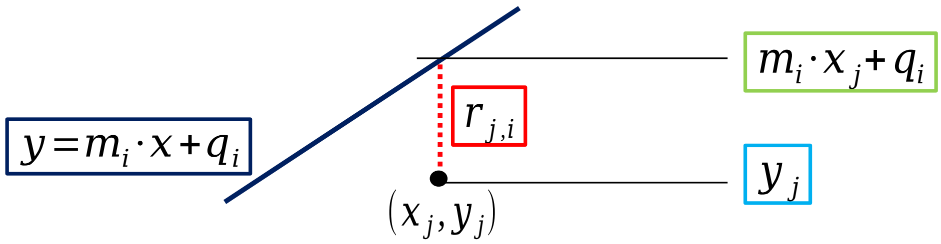
- For a line fitting we can define residuals as

$$\underbrace{r_{i,j}}_{\text{Distance between data}_j\text{ and model}_i} = \underbrace{y_j}_{\text{Real value of data}_j} - \underbrace{(m_i \cdot x_j + q_i)}_{\text{Estimated value for data}_j\text{ given by model}_i}$$



Per esempio posso usare come residuo la differenza di ordinata. Ok non sarà il residuo migliore immaginabile ma è semplice da calcolarsi

LS Line Fitting

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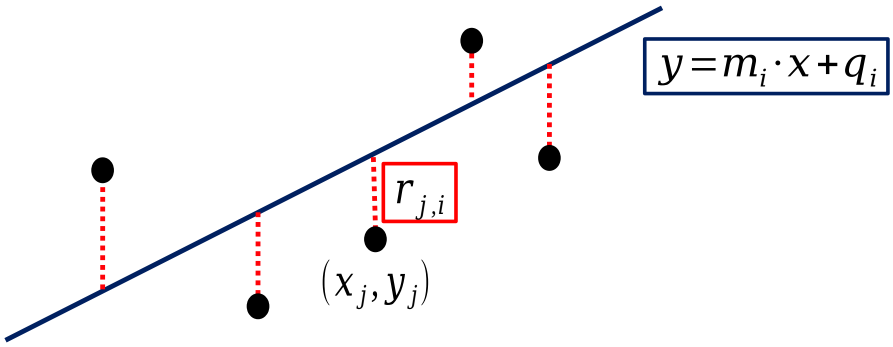
- Given Error Function definition:

$$E(model_i, data) = \sum_{j=1}^{n} r_{i,j}^2 = \sum_{j=1}^{n} \left[y_j - (m_i \cdot x_j + q_i) \right]^2$$



LS line fitting

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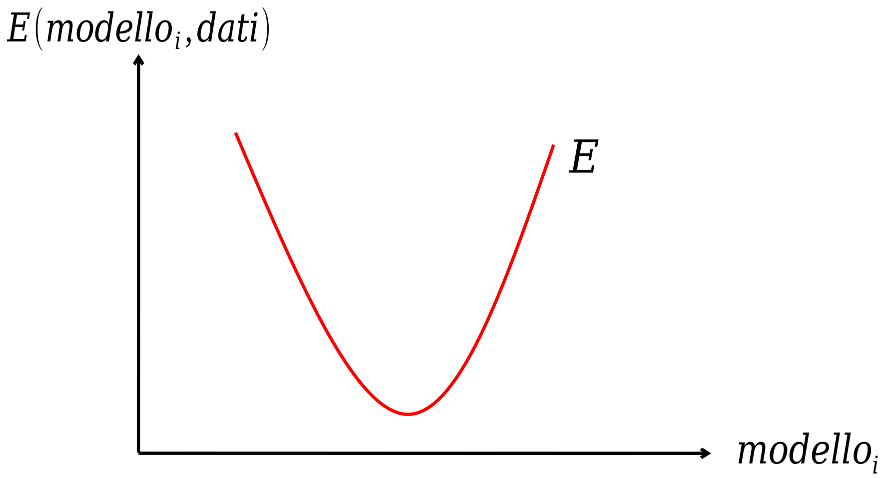
- Choose a model and its parameters
- Define a error function E(model$_i$, data) to evaluate model$_i$
- Adapt model to data
  - find the parameters set that minimize E()

LS Line Fitting

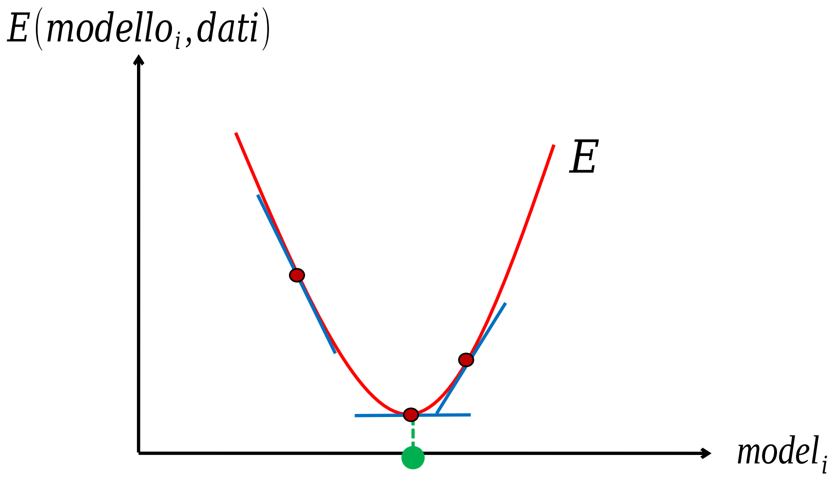
- Consider E() graph



Somma di QUADRATI ⟶ quindi andamento a PARABOLA
Minimizzo...

LS Line Fitting

- We need to find the parameter set that minimize E()

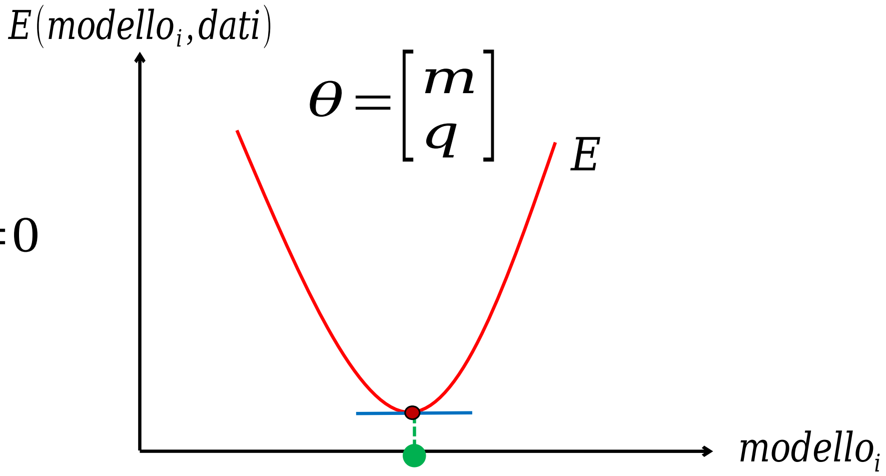


Come minimizzo? Derivata no? Dove la derivata è zero avrò il minimo

LS Line Fitting

- Compute gradient and try to obtain 0

$$\frac{\partial E}{\partial \theta} = \begin{bmatrix} \frac{\partial E}{\partial m} \\ \frac{\partial E}{\partial q} \end{bmatrix} = 0$$



Sostanzialmente devo azzerare la derivata di E(m,q) sia per m che per q

LS Line Fitting

- Partial derivative of E with respect to m

$$\frac{\partial E}{\partial m} = \frac{\partial}{\partial m} \sum_{j=1}^{n} \left(y_j - (m x_j + q)\right)^2 = 0$$

$$.= \frac{\partial}{\partial m} \sum_{j=1}^{n} \left(y_j^2 - 2 y_j (x_j m + q) + (m x_j + q)^2\right)$$

$$.= \sum_{j=1}^{n} \frac{\partial}{\partial m} \left(y_j^2 - 2 x_j y_j m - 2 y_j q + x_j^2 m^2 + 2 x_j q m + q^2\right)$$

$$.= \sum_{j=1}^{n} \left(2 x_j^2 m - 2 x_j y_j + 2 x_j q\right)$$

LS Line Fitting

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- Partial derivative of E with respect to m

$$\frac{\partial E}{\partial m} = \sum_{j=1}^{n} \left(2 x_j^2 m - 2 x_j y_j + 2 x_j q \right) = 0$$

$$\left(\sum_{j=1}^{n} x_j^2 \right) m + \left(\sum_{j=1}^{n} x_j \right) q = \sum_{j=1}^{n} x_j y_j \qquad \textbf{(Eq. 1)}$$

LS Line Fitting

- Partial derivative of E with respect to q

$$\frac{\partial E}{\partial q}=\frac{\partial}{\partial q}\sum_{j=1}^{n}(y_j-(mx_j+q))^2=0$$

$$.=\frac{\partial}{\partial q}\sum_{j=1}^{n}\left(y_j^2-2y_j(x_jm+q)+(mx_j+q)^2\right)$$

$$.=\sum_{j=1}^{n}\frac{\partial}{\partial q}\left(y_j^2-2x_jy_jm-2y_jq+x_j^2m^2+2x_jqm+q^2\right)$$

$$.=\sum_{j=1}^{n}\left(2x_j^2m+2q-2y_j\right)$$

Il procedimento è ovviamente identico al precedente se non la derivata fatto salvo l’ultima riga in cui infatti applico la derivata

LS Line Fitting

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- Partial derivative of E with respect to q

$$\frac{\partial E}{\partial q} = \sum_{j=1}^{n} \left(2 x_j m + 2q - 2 y_j\right) = 0$$

$$\left(\sum_{j=1}^{n} x_j\right) m + \underbrace{\sum_{j=1}^{n} 1}_{n} q = \left(\sum_{j=1}^{n} x_j\right) m + nq = \sum_{j=1}^{n} y_j \qquad \textbf{(Eq. 2)}$$

Il procedimento è ovviamente identico al precedente se non la derivata fatto salvo l'ultima riga in cui infatti applico la derivata

LS Line Fitting

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- We can write equations 1 & 2 in a matrix form

$$\begin{bmatrix} \sum x_j^2 & \sum x_j \\ \sum x_j & n \end{bmatrix} \begin{bmatrix} m \\ q \end{bmatrix} = \begin{bmatrix} \sum x_j y_j \\ \sum y_j \end{bmatrix}$$

LS Line Fitting

$$X = \begin{bmatrix} x_1 & 1 \\ \vdots & \vdots \\ x_n & 1 \end{bmatrix} \qquad Y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} \qquad \theta = \begin{bmatrix} m \\ q \end{bmatrix}$$

LS Line Fitting

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- We can see previous equation as:

$$\underbrace{\begin{bmatrix} \sum x_j^2 & \sum x_j \\ \sum x_j & n \end{bmatrix}}_{X^T X} \underbrace{\begin{bmatrix} m \\ q \end{bmatrix}}_{\theta} = \underbrace{\begin{bmatrix} \sum x_j y_j \\ \sum y_j \end{bmatrix}}_{X^T Y}$$

LS Line Fitting

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- We can then solve this as

$$X^T X \theta = X^T Y$$

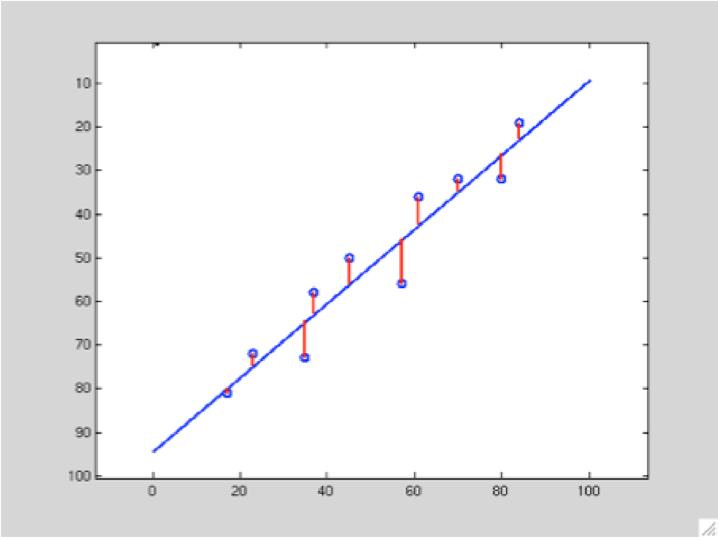
$$\Downarrow$$

$$\theta = \left(X^T X\right)^{-1} X^T Y$$

θ is the unknown!

I vari X e Y rammento che sono TERMINI NOTI. Quindi a dispetto del numero di parametri coinvolti il calcolo è relativamente semplice

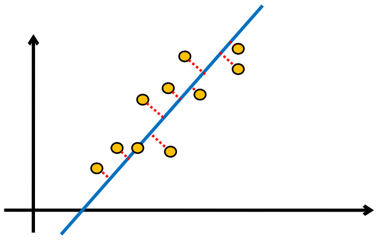
LS Line Fitting



Esempio di risultato

LS Line Fitting – algebraic distance

- (m,q) are not always the best parameter choiche
  - Vertical lines?
- A more general approach is $ax+by+c=0$
- We can also use algebraic distance as estimator for fitting



LA differenza sulla Y è molto semplice ma forse posso pensare di usare un metodo migliore. Per esempio acambio modello e uso la distanza come funzione di errore

LS Line Fitting – algebraic distance

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- Cost function E can now be the actual distance

$$E_i = \sum_{j=1}^{n} \left(a_i x_j + b_i y_j + c_i\right)^2$$

- Let's find the parameter set that minimize E_i

Non è esattamente la distanza algebrica occorrerebbe dividere per la radice di a^2+b^2, ma non sottilizziamo

LS Line Fitting – algebraic distance

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- Partial derivatives can be written as

$$\frac{\partial E}{\partial a}=\sum_{j=1}^{n}\left(2x_j^2a+2x_jy_jb+2x_jc\right)$$

$$\frac{\partial E}{\partial b}=\sum_{j=1}^{n}\left(2x_jy_ja+2y_j^2b+2y_jc\right)$$

$$\frac{\partial E}{\partial c}=\sum_{j=1}^{n}\left(2x_ja+2y_jb+2c\right)$$

Come prima derivo. A, b e c sono le incognite!

LS Line Fitting – algebraic distance

- Again, put them as homogeneous equation

$$\underbrace{\begin{bmatrix} \sum x_j^2 & \sum x_j y_j & \sum x_j \\ \sum x_j y_j & \sum y_j^2 & \sum y_j \\ \sum x_j & \sum y_j & n \end{bmatrix}}_{A} \cdot \underbrace{\begin{bmatrix} a \\ b \\ c \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}}_{0}$$

LS Line Fitting – algebraic distance

- Then we should solve the homogeneous equation

$$A \cdot x = 0$$

- There is always a solution?
  - No!
- Moreover, ill posed when A $\in$ $R_{3\times 3}$ has a rank<3

Ovvio, il rango mi dice quante linee sono linearmente indipendenti se inferiore dimensione $\longrightarrow$ infinite soluzioni

Ma in generale sarà superiore. Anche perché se ho 1 punto ho realmente infinite soluzioni, se ne ho due la calcolo banalmente.

LS Line Fitting – algebraic distance

- We can use (again) SVD for a minimization

$$A = UDV^T$$

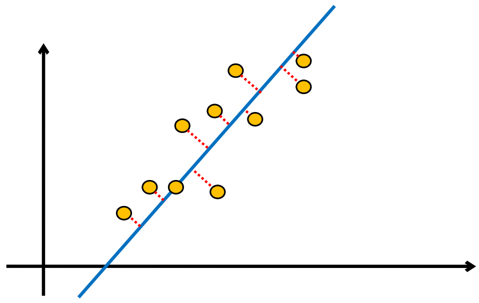
- x is the last column of V

Ancora Singular Value Decomposition

Rammento che è semplice quanto il precedente. Sia OpenCV che Matlab forniscono SVD

LS Summary

- Summary
  - Simple approach
    - n ≤ 2 ⟶ closed form
    - n > 2 ⟶ SVD
  - No thresholds
  - Results is good when until we have Gaussian noise…

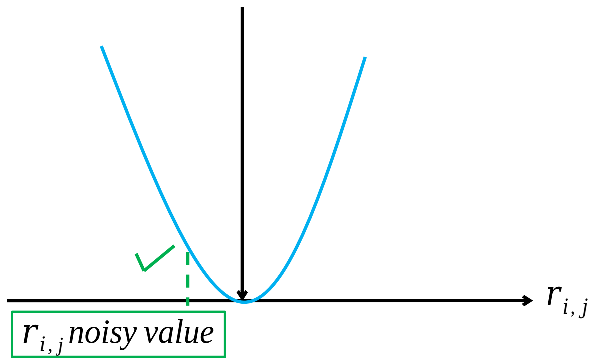


Il fatto di non aver soglie è vero anche nel caso precedente.

LS issue: outliers

- Not always Gaussian noise
- LS is highly affected by outliers

$$f(r_{i,j}) \propto r_{i,j}^2$$



Quanto funziona bene questo sistema? Se il rumore è gaussiano sarà piú o meno centrato attorno al valore corretto $\longrightarrow$ risultati ragionevoli

Ma rammanto che l'andamento è parabolico visto la scelta di usare dei quadrati.

Il peso cambia significativamente con la distanza quindi!

LS issue: outliers

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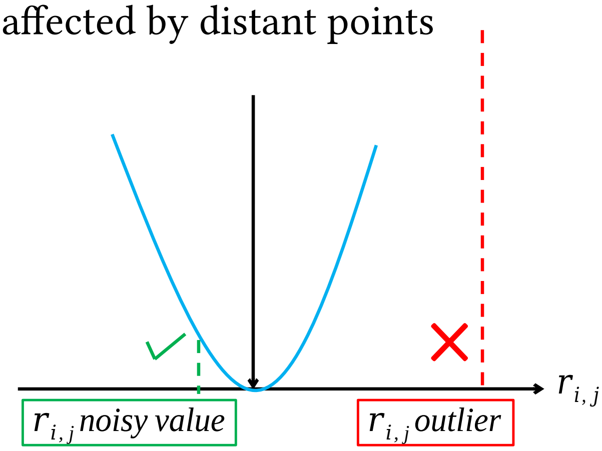
- Not always Gaussian noise
- LS is highly affected by distant points

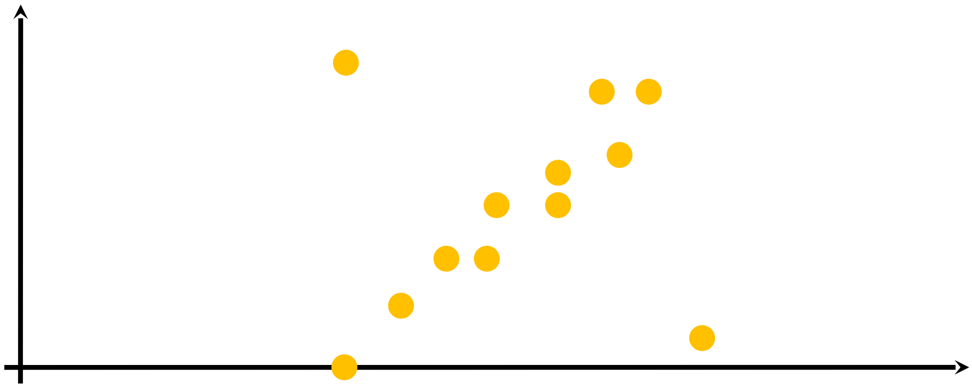
$$f(r_{i,j}) \propto r_{i,j}^2$$



Se ho punti lontani proprio completamente sbagliati questi mi forniscono un contributo enorme al risultato facendomi sballare il tutto

LS issue: outliers

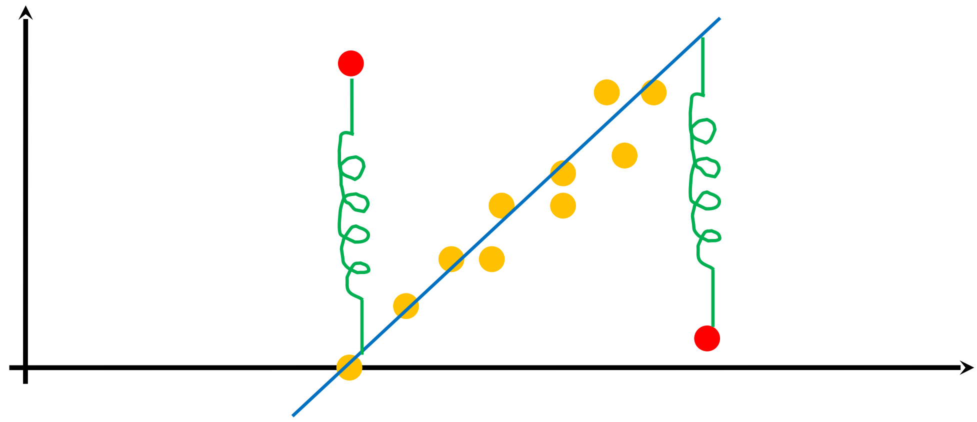
- Example: what is the right fitting in this case?



Si può notare come ci siano due punti decisamente non allieneati

LS issue: outliers

- Example: what is the right fitting in this case?



Il fitting realistico sarebbe quello indicato dalla retta, ma quei due punti mi fanno sballare tutto

LS issue: outliers

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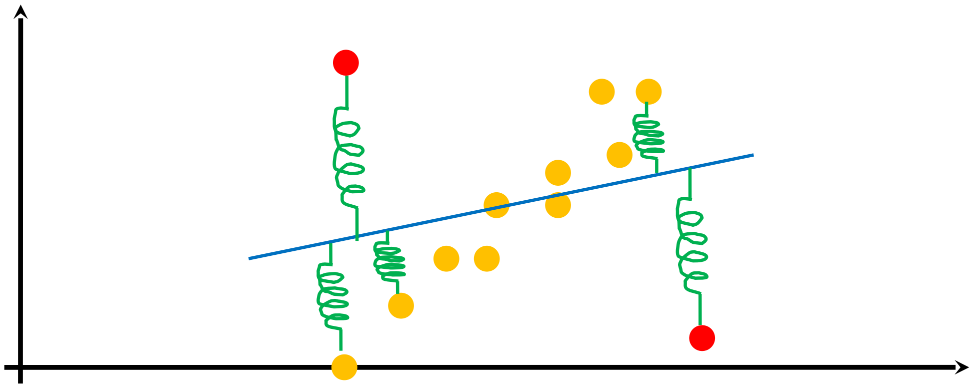
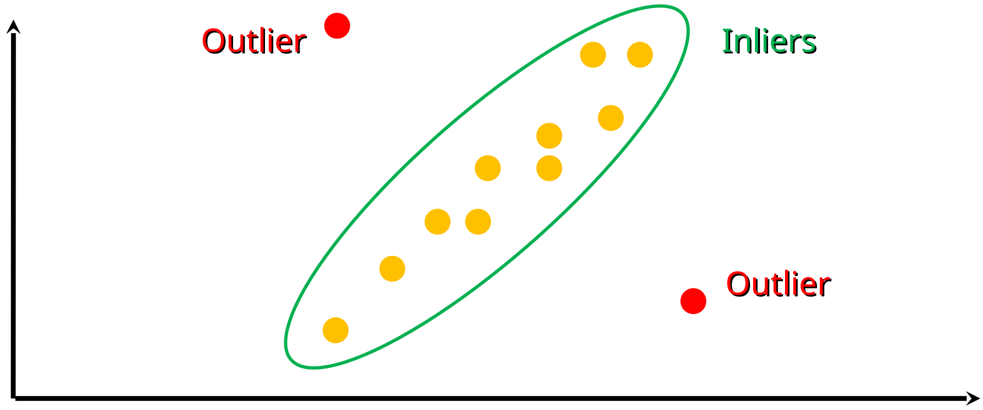
- Example: what is the right fitting in this case?



Quindi il risultato sarà una cosa del genere

Outliers

- Inliers: data which belong to the model
- Outliers: data not belonging to the model



Di fatto potrei dividere i dati in due classi: gli inlier che sono quelli che vorrei considerare e gli outlier che sono quelli che dovrei scartare...

Ma il metodo dei minimi quadrati come lo abbiamo visto non ci permette di farlo (almeno da solo)

Robust Statistics

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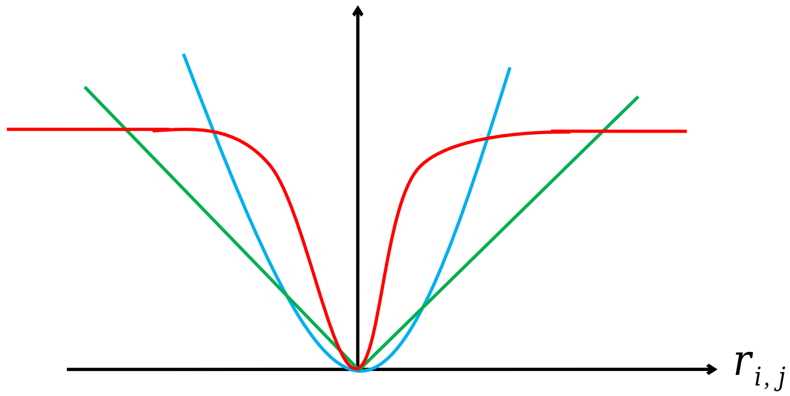
- Underlying idea: modify data weight depending on their distance from the model

$$f(r_{i,j}) \propto r_{i,j}^2$$

$$f(r_{i,j}) \propto |r_{i,j}|$$

$$f(r_{i,j}) \propto \frac{r_{i,j}^2}{r_{i,j}^2+\sigma^2}$$



Un primo approccio potrebbe essere quello di modificare la funzione E(). Ad esempio non usando i quadrati oppure usandoli ma portando a saturazione la funzione oltre una certa distanza.

Nessuna di queste soluzioni è in realtà completamente risolutiva in quanto di fatto gli outlier continuano a contribuire

Robust Statistics

- Instead of squared residuals sum we use a function ρ()

$$min \sum_{j=1}^{n} r_i^2 \quad \rightarrow \quad min \sum_{j=1}^{n} \rho(r_i)$$

- The function is symmetrical and features a minimum in zero
  - Loss function

In sostanza passo dai quadrati ad altra funzione. Chiaro che devo ricalcolarmi tutto quanto fatto con i quadrati (ma non lo vediamo)

Robust Statistics RANSAC

- Estimation can also divided in two steps
  - Classify data among Inliers and Outliers
  - Generate a model using inliers only
- RANSAC (RANdom SAmple Consensus) is a widely used approach
  - M. A. Fischler and R. C. Bolles (June 1981). *"Random Sample Consensus: A Paradigm for Model Fitting with Applications to Image Analysis and Automated Cartography"*. Comm. of the ACM 24: 381--395

RANSAC usa un approccio interessante ed è tuttora molto usato.

L'idea di base è quella di cercare di scartare gli outliers.

Anche con percentuali di outliers significative funziona molto bene.

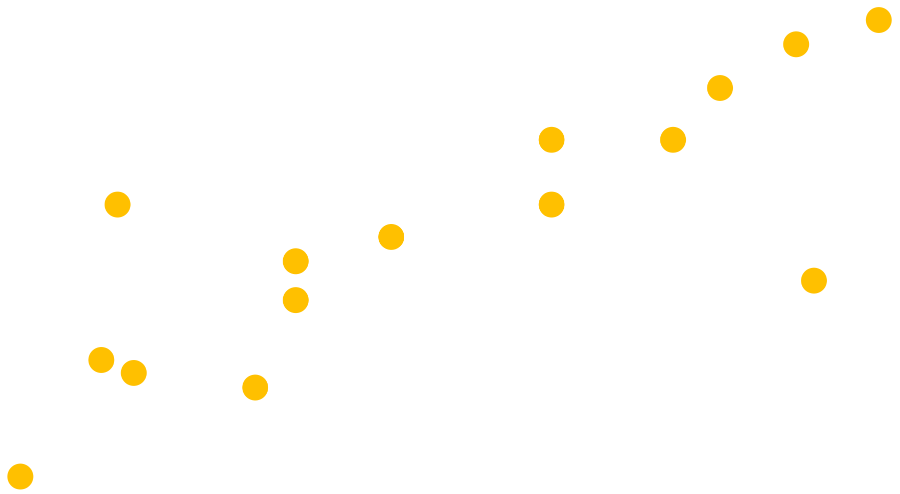
RANSAC

- Given a set of points
  - $(x_1, y_1), (x_2,y_2), \ldots, (x_n,y_n)$
  - Potentially including outliers…
- Estimate parameters
  - Example: $y=mx+q$ or $ax+by+c=0$

Adottiamo lo stesso problema visto prima, fitting di una retta.

RANSAC line fitting

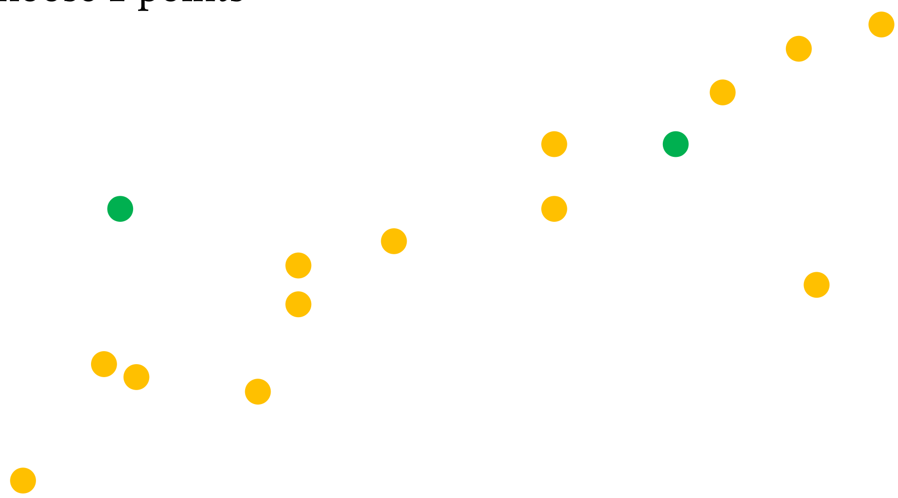
- We have some outliers...



Con qualche outleier

RANSAC line fitting

- Randomly choose 2 points



L'idea è selezionare coppie di punti (coppie perché una coppia mi definisce una retta, in caso di altri modelli il numero di dati selezionati potrebbe essere maggiore) e valutare la retta che ne deriva

RANSAC line fitting

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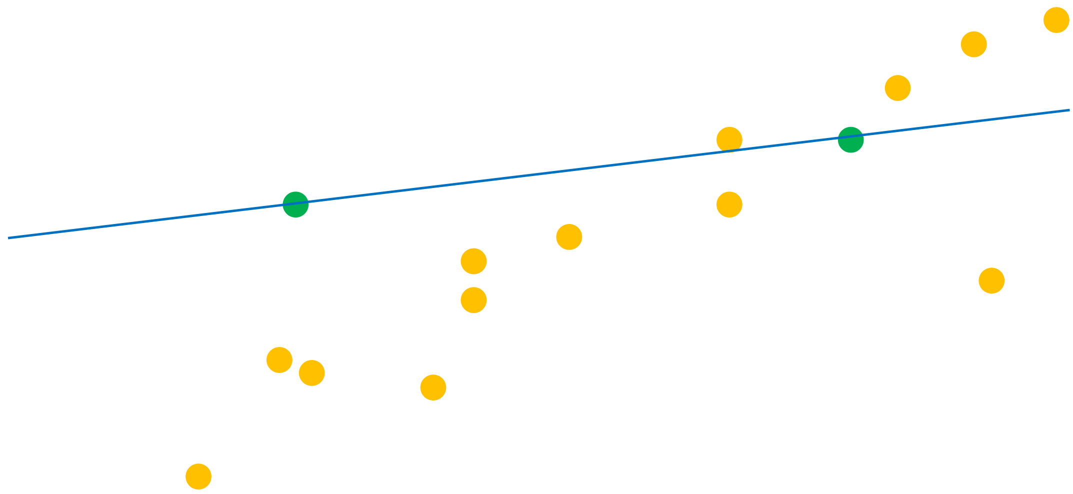
- Estimate model



RANSAC line fitting

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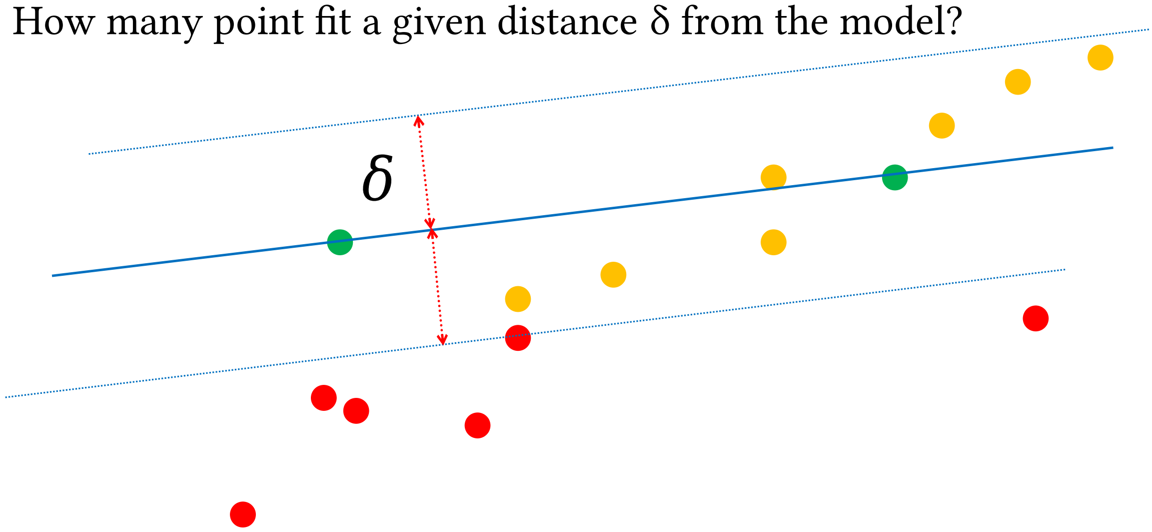
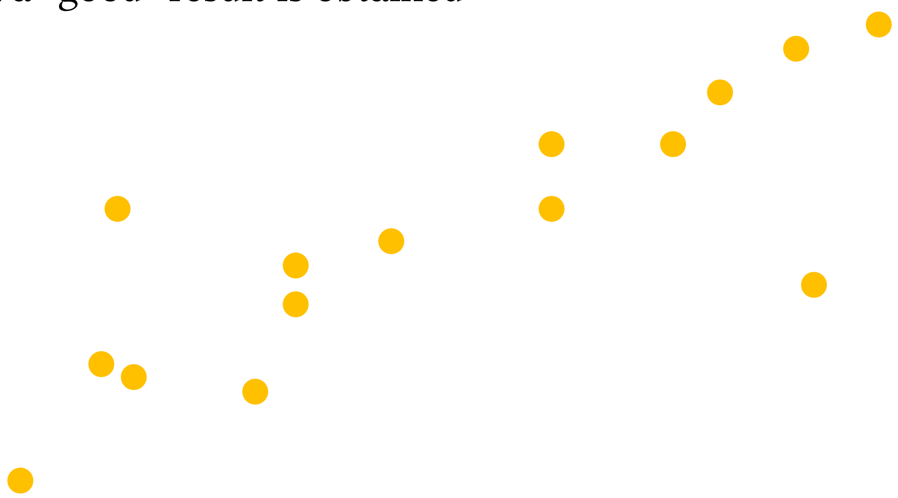
- How many point fit a given distance δ from the model?



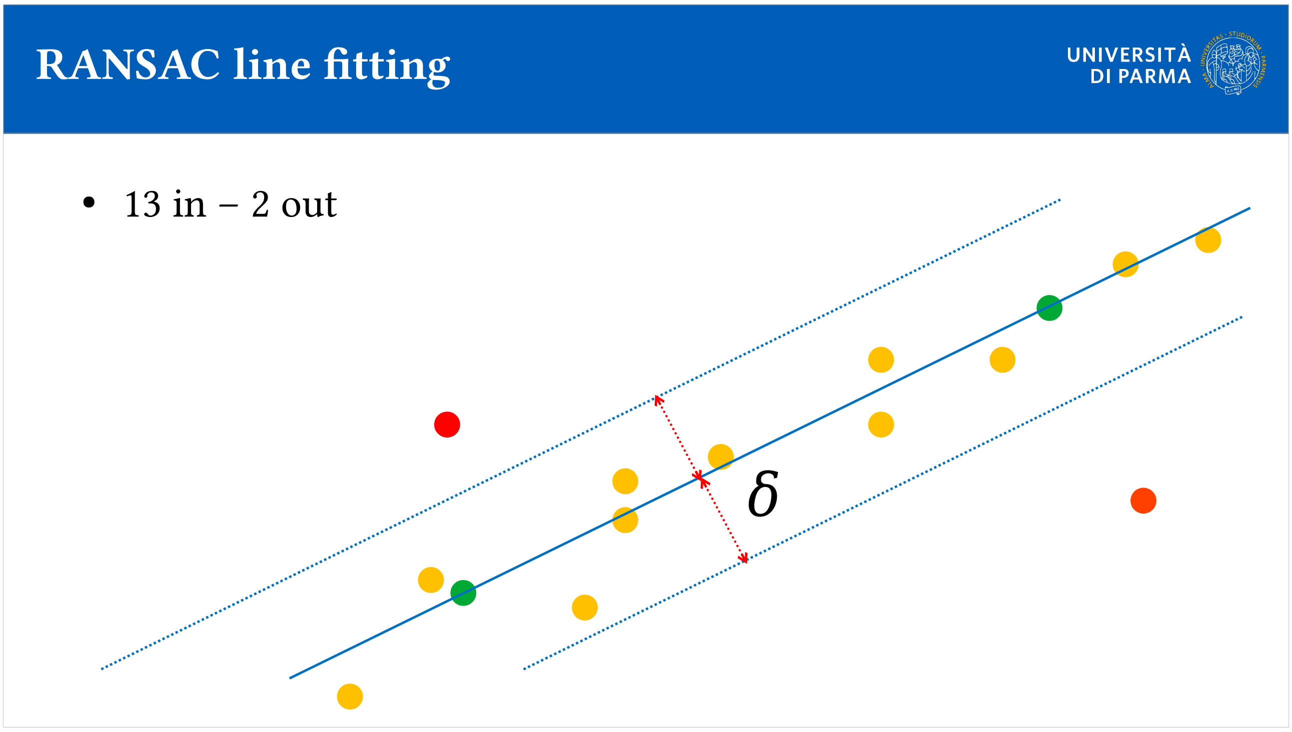
Come la valutiamo? Consideriamo un intorno di tale retta e guardiamo quanti altri punti ci cadono dentro e ce lo annotiamo

RANSAC line fitting

- Repeat until a “good” result is obtained



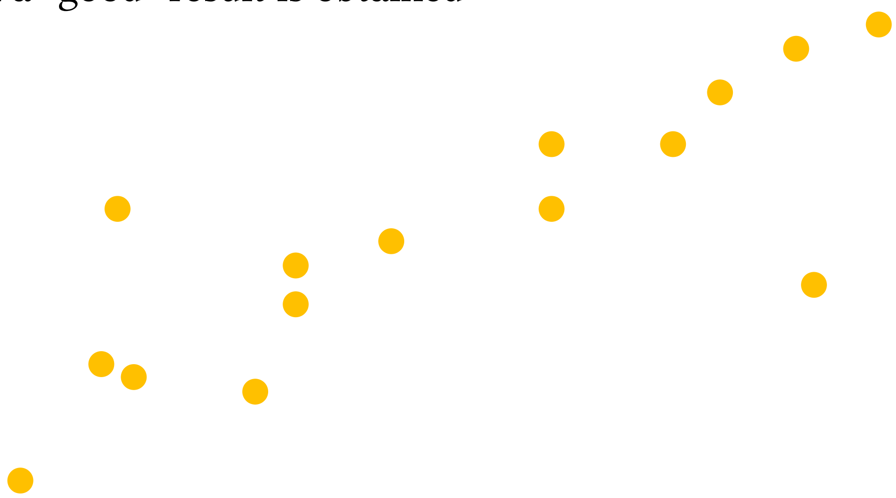
Perché poi proveremo altri due punti



In questo caso il risultato è migliore in quanto massimizzo il numero di inlier

RANSAC line fitting

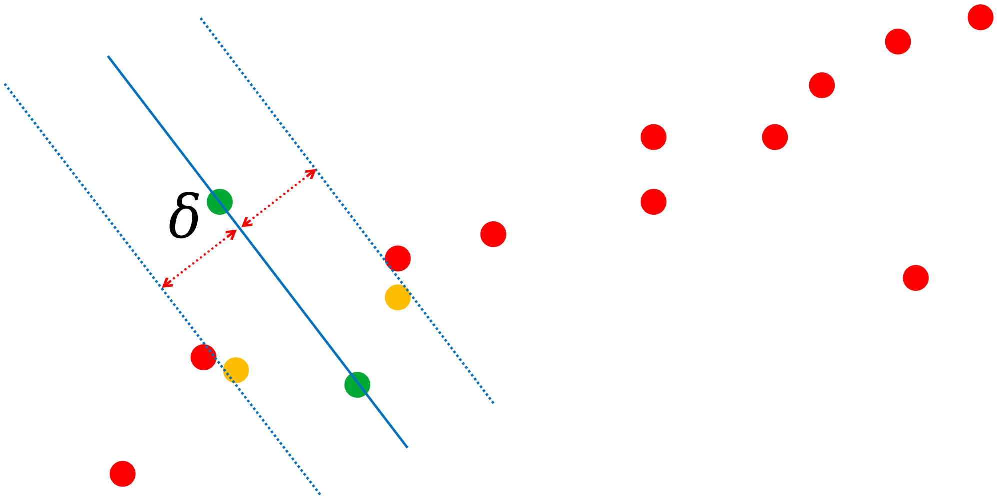
- Repeat until a “good” result is obtained



Però devo ripetere molte volte...

RANSAC line fitting

- 4 in…



How many iterations?

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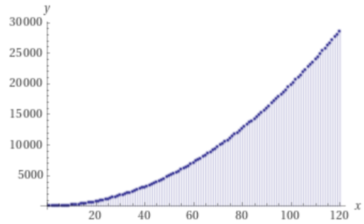
- Let’s just consider the straight line example
- We have n=15 points
- In order to test a line we need s=2 points

$$\binom{n}{s} = \frac{n!}{(n-s)!\,s!} = 105$$



Fino a che il risultato non è buono (sì ma cosa significa?)

Nel caso in esame ho 15 punti, le possibilità di coppie distinte le posso calcolare con quella formula.

Devo fare 105 confronti? Che poi quel numero come capite esplode al crescere di n (1500 ⟶ ~1 M, 150 ⟶ ~11.000)

In realtà posso fermarmi molto prima. Vediamo come e perché

How many iterations?

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- Given:
  - e ⟶ probability a given point is an outlier
  - s ⟶ number of points in a subset
  - N ⟶ number of subsets (the unknown)
  - p ⟶ probability to have at least 1 good subset

$$p = 1 - \left(1 - (1-e)^s\right)^N$$

Affinché il tutto funzioni devo essere sicuro di “pescare” almeno un buon set (nel nostro esempio, una coppia buona di punti).

Quella formula (provate a scomporla dalla parentesi interna a salire, come nelle slide successive) mi dice qual è la probabilità di pescare un buon set di s elementi con N pescate.

Se ci pensiamo:

- S è noto (nel nostro caso 2 ma ovviamente dipende caso per caso)
- E anche (come anche? Davvero so quanti outlier ci sono in percentuale? No, non lo so ma posso fare una stima. Basta essere sufficientemente cautelativi, mal che vada si riflette su N aumentandolo a parità di p)
- P anche lui è noto. È quello che voglio ottenere. Quindi lo stabilisco a priori. E tipicamente ad un valore alto, del tipo 99%

Cosa non è noto? Il numero di pescate N necessario per essere ragionevolmente sicuri di otteneerne una buona.

How many iterations?

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- Given:
  - e ⟶ probability a given point is an outlier
  - s ⟶ number of points in a subset
  - N ⟶ number of subsets (the unknown)
  - p ⟶ probability to have at least 1 good subset

$$p = 1 - \left(1 - \underbrace{(1-e)}_{\text{Inlier Probability}}^{s}\right)^{N}$$

How many iterations?

- Given:
  - e ⟶ probability a given point is an outlier
  - s ⟶ number of points in a subset
  - N ⟶ number of subsets (the unknown)
  - p ⟶ probability to have at least 1 good subset

$$p = 1 - \left(1 - \underbrace{\left(1 - e\right)^{s}}_{\text{All Inlier Probability in the Subset}}\right)^{N}$$

How many iterations?

- Given:
  - e ⟶ probability a given point is an outlier
  - s ⟶ number of points in a subset
  - N ⟶ number of subsets (the unknown)
  - p ⟶ probability to have at least 1 good subset

$$p=1-\left(1-\left(1-e\right)^{s}\right)^{N}$$

One or more Outliers Probability in the Subset

How many iterations?

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- Given:
  - e ⟶ probability a given point is an outlier
  - s ⟶ number of points in a subset
  - N ⟶ number of subsets (the unknown)
  - p ⟶ probability to have at least 1 good subset

$$p = 1 - \left(1 - (1 - e)^s\right)^N$$

Probability that N subsets always contain outliers

How many iterations?

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- Given:
  - e ⟶ probability a given point is an outlier
  - s ⟶ number of points in a subset
  - N ⟶ number of subsets (the unknown)
  - p ⟶ probability to have at least 1 good subset

$$p = 1 - \left(1 - \left(1 - e\right)^s\right)^N$$

Probability to have at least one uncontaminated subset

How many iterations?

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- N can be then computed as f(p,e)
  - s is known given the model

$$\left(1-\left(1-e\right)^{s}\right)^{N}=p-1 \quad \rightarrow \quad N=\frac{\log\left(1-p\right)}{\log\left(1-\left(1-e\right)^{s}\right)}$$

Fissato p il mio N lo ricavo con quella formula

How many iterations?

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- Assuming we need a “strong” probability (i.e. p=0.99)

$$N = \frac{\log(1-p)}{\log(1-(1-e)^s)}$$

	proportion of outliers e						
s	5%	10%	20%	25%	30%	40%	50%
2	2	3	5	6	7	11	17
3	3	4	7	9	11	19	35
4	3	5	9	13	17	34	72
5	4	6	12	17	26	57	146
6	4	7	16	24	37	97	293
7	4	8	20	33	54	163	588
8	5	9	26	44	78	272	1177

Guardiamo ad esempio il nostro caso (la prima riga)

Anche con un numero di outlier decisamente alto (50%) mi bastano 17 tenmtativi a caso per avere una probabilità del 99% di ottenere la combinazione giusta.

How many iterations (N)?

- Again let's consider the straight line example
- We have n=15 points
- We also estimate a ~20% outliers
- In order to test a line we need s=2 points
- We want a p=0.99

$$N = 5 \ll 105$$

Questo è un esempio piú realistico

How many iterations?

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- Therefore it is not necessary to test all subsets
- We can simply randomly choose N
  - N is usually << all possible combinations
- We can further reduce N
  - Stop iterations when a "sufficient" number of inliers is found

$$T = (1 - e) \cdot n$$

In genere, visto che vorrei la sicurezza si usano N ancor piú grandi. MA comunque molto piú piccoli del massimo numero di combinazioni.

Visto che io fisso la probabilità di outlier ad e, quel T (con n numero di punti), mi rappresenta quanti inlier ho. Ma se arrivo ad un tentativo in cui trovo T o anche piú inlier direi che mi posso fermare.

RANSAC: improvements

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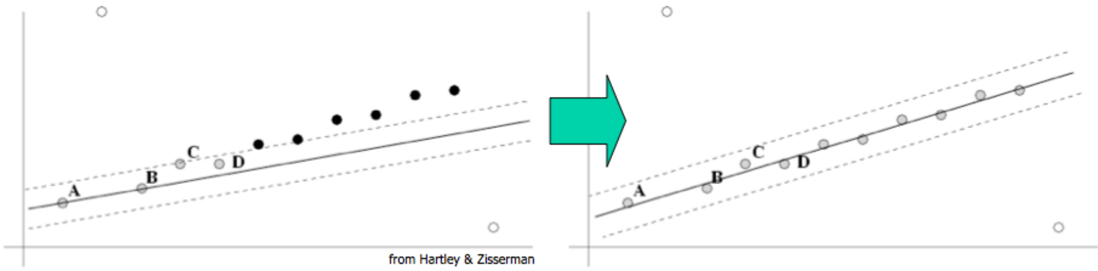
- Once a model is tested it can be refined considering all inliers
  - LS can be used again



Passata questa prima fase io ho eliminato degli outlier. Ho sicuramente trovato un modello ragionevole ma posso anche raffinarlo usando i soli inlier e uno dei modi già visti (qui LS dovrebbe funzionare comunque bene visto che non ho piú outlier)

RANSAC: pro&cons

- It often works!
- We need data assumptions
  - inliers/outliers probability
- When several outliers it may be slow...
  - No upper time
  - You can limit iterations $\longrightarrow$ far from optimal solution

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Model Fitting

Question time!

Model Fitting